CRIME AWARE: PREDICTIVE CRIME ZONE WITH MAPPING OPTIMIZATIONAND SAFETY ALERTS FOR WOMEN IN HIGH-RISK ALERTS

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Supervisor　　　　Head of the Department

Submitted to project and viva examination held on……………………………

INTERNAL EXAMINER　　　　EXTERNAL EXAMINER

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Abstract

Women in crime hotspots are at heightened risk of facing various criminal activities, including sexual harassment, assault, domestic violence, stalking, and human trafficking. For instance, in 2025, a woman in Bengaluru was kidnapped in a high-crime area, highlighting the urgent need for proactive safety measures. Identifying these high-risk areas is essential for effective law enforcement and personal safety planning. Traditional crime prediction methods often rely on static historical data and simple statistical models, which fail to capture complex spatial and temporal dependencies in criminal activity, leading to inaccurate predictions and delayed interventions. To address these limitations, this project introduces a predictive system that leverages Spatio-Temporal Graph Convolutional Networks (ST-GCN) to identify and map crime hotspots in real time. The system continuously analyzes crime trends, considering both historical and live incident data, enabling dynamic hotspot detection. When a woman uses the system, she can select her source and destination, and the system evaluates multiple possible routes, highlighting hotspot-prone paths and suggesting safe alternatives. Additionally, the system can provide estimated travel times and alert the user about areas with recent incidents along the route. By recommending the safest path and visualizing potential risks, the platform empowers women to make informed travel decisions. The system also supports alert notifications to local authorities in case of unusual or emerging risks along a selected route, enabling rapid preventive action. By integrating spatio-temporal predictive modeling, route analysis, and real-time notifications, this approach improves personal safety, assists law enforcement in proactive monitoring, and contributes to the creation of safer urban environments for women. Ultimately, the project promotes informed decision-making, risk awareness, and confidence while navigating public spaces.

சுருக்கம்

குற்றச் சம்பவங்கள் அதிகம் நடக்கும் பகுதிகளில் உள்ள பெண்கள், பாலியல் துன்புறுத்தல், தாக்குதல், குடும்ப வன்முறை, பின்தொடர்தல் மற்றும் ஆட்கடத்தல் உள்ளிட்ட பல்வேறு குற்றச் செயல்களை எதிர்கொள்ளும் அதிக அபாயத்தில் உள்ளனர். உதாரணமாக, 2025-ஆம் ஆண்டில், பெங்களூருவில் அதிக குற்றங்கள் நடக்கும் பகுதியில் ஒரு பெண் கடத்தப்பட்டார். இது, முன்கூட்டியே பாதுகாப்பு நடவடிக்கைகளை எடுக்க வேண்டியதன் அவசரத் தேவையை எடுத்துக்காட்டுகிறது. திறம்பட்ட சட்ட அமலாக்கம் மற்றும் தனிநபர் பாதுகாப்புத் திட்டமிடலுக்கு, இந்த அதிக அபாயமுள்ள பகுதிகளைக் கண்டறிவது அவசியமாகும். பாரம்பரிய குற்றக் கணிப்பு முறைகள் பெரும்பாலும் நிலையான வரலாற்றுத் தரவுகள் மற்றும் எளிய புள்ளியியல் மாதிரிகளைச் சார்ந்துள்ளன. இவை, குற்றச் செயல்களில் உள்ள சிக்கலான இடஞ்சார்ந்த மற்றும் காலஞ்சார்ந்த சார்புகளைக் கண்டறியத் தவறுவதால், தவறான கணிப்புகளுக்கும் தாமதமான தலையீடுகளுக்கும் வழிவகுக்கின்றன. இந்த வரம்புகளை நிவர்த்தி செய்ய, இந்தத் திட்டம், இடஞ்சார்ந்த-காலஞ்சார்ந்த வரைபட சுருள்வலையமைப்புகளை (ST-GCN) பயன்படுத்தி, நிகழ்நேரத்தில் குற்றச் சம்பவங்கள் அதிகம் நடக்கும் பகுதிகளைக் கண்டறிந்து வரைபடமாக்கும் ஒரு கணிப்பு அமைப்பை அறிமுகப்படுத்துகிறது. இந்த அமைப்பு, வரலாற்று மற்றும் நிகழ்நேர சம்பவத் தரவுகளைக் கருத்தில் கொண்டு, குற்றப் போக்குகளைத் தொடர்ச்சியாகப் பகுப்பாய்வு செய்து, மாறும் தன்மையுடன் குற்றச் சம்பவங்கள் அதிகம் நடக்கும் பகுதிகளைக் கண்டறிய உதவுகிறது. ஒரு பெண் இந்த அமைப்பைப் பயன்படுத்தும்போது, அவர் தனது புறப்படும் இடம் மற்றும் சேருமிடத்தைத் தேர்ந்தெடுக்கலாம். இந்த அமைப்பு பல சாத்தியமான வழிகளை மதிப்பீடு செய்து, அபாயப் பகுதிகள் அதிகம் உள்ள பாதைகளைக் குறிப்பிட்டு, பாதுகாப்பான மாற்று வழிகளையும் பரிந்துரைக்கிறது. மேலும், இந்த அமைப்பு உத்தேச பயண நேரங்களை

வழங்குவதோடு, பயண வழியில் சமீபத்தில் சம்பவங்கள் நடந்த பகுதிகள் குறித்தும் பயனரை எச்சரிக்கிறது. மிகவும் பாதுகாப்பான பாதையைப் பரிந்துரைப்பதன் மூலமும், ஏற்படக்கூடிய அபாயங்களைக் காட்சிப்படுத்துவதன் மூலமும், இந்தத் தளம் பெண்கள் தகவலறிந்த பயண முடிவுகளை எடுக்க உதவுகிறது. தேர்ந்தெடுக்கப்பட்ட ஒரு வழியில் அசாதாரணமான அல்லது புதிதாக உருவாகும் அபாயங்கள் ஏற்பட்டால், உள்ளூர் அதிகாரிகளுக்கு எச்சரிக்கை அறிவிப்புகளை வழங்குவதையும் இந்த அமைப்பு ஆதரிக்கிறது, இது விரைவான தடுப்பு நடவடிக்கைகளை எடுக்க உதவுகிறது. இட-கால முன்கணிப்பு மாதிரியாக்கம், வழித்தடப் பகுப்பாய்வு மற்றும் நிகழ்நேர அறிவிப்புகளை ஒருங்கிணைப்பதன் மூலம், இந்த அணுகுமுறை தனிப்பட்ட பாதுகாப்பை மேம்படுத்துகிறது, சட்ட அமலாக்கத் துறைக்கு முன்கூட்டியே கண்காணிக்க உதவுகிறது, மேலும் பெண்களுக்குப் பாதுகாப்பான நகர்ப்புறச் சூழல்களை உருவாக்குவதற்கும் பங்களிக்கிறது. இறுதியாக, இந்தத் திட்டம் பொது இடங்களில் பயணிக்கும்போது தகவலறிந்த முடிவெடுத்தல், அபாய விழிப்புணர்வு மற்றும் தன்னம்பிக்கையை ஊக்குவிக்கிறது.

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CHAPTER 1

INTRODUCTION

1.1. OVERVIEW

Crimes against women are of various types as crimes involving sex for economic gains including prostitution, wrongful confinement, trafficking, dowry extortion, rape, assault, harassment at work place, gang-rape, acid-attack, kidnapping, and other immoral acts are injurious to the society. In present scenario cases of murder, rape, molestation, sexual abuse, and eve-teasing etc. Women feel discrimination, harassment and domestic violence etc at work. Use of money and muscle power to save accused. Politics in the name of caste, religion etc take benefit as well. India at 108th place according to Global Gender Gap Index 2017. A 2017 report by Global Peace Index had claimed India to be the fourth most dangerous country for women travellers.

1.1. Crime

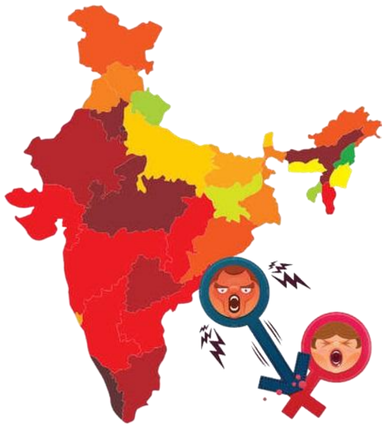
Violence against women and girls is rooted in gender-based discrimination and social norms. Women are so helpless in the Indian society where many female goddesses are worshipped but still people don't respect women. It is high time when the social revolution is needed to root out this evil from Indian society and given respect and recognition to our women.

India's Crime Statistics on Violence Against Women

Overall, 60.96 lakh crimes were registered in the country last year, a decline of 7.6% from 2020 For the fifth consecutive year, Assam recorded the highest rate of crimes against women in 2021, according to the National Crime Record Bureau's "Crime in India" report for last year. The rate is calculated as the number of crimes recorded for every one lakh people. According to the data, the rate of crimes against women last year stood at 168.3 in Assam, an increase from 154.3 in 2020. Next on the list were Odisha at 137.8, Haryana at 119.7, and Rajasthan at 105.4. In terms of the actual number of cases, Uttar Pradesh with 56,083 cases, topped the list, followed by Rajasthan (40,738), Maharashtra (39,526), West Bengal (35,884) and Odisha (31,352). Nagaland reported the lowest crime rate against women in 2021 (5.5). It also had the lowest number of actual cases – 54. A total of 4,28,278 cases of crimes against women were registered in the country under the Indian Penal Code and special and local laws last year. The number is 15.3% more than the 3,71,503 cases registered in 2020.

Crime Hotspot

Crime hotspots are areas that have high crime intensity. These are usually visualized using a map. They are developed for researchers and analysts to examine geographic areas in relation to crime. Hot spots are areas with a high occurrence of crime. These areas can be anywhere. They can be bars, malls, neighbourhoods, pretty much anywhere criminals target.



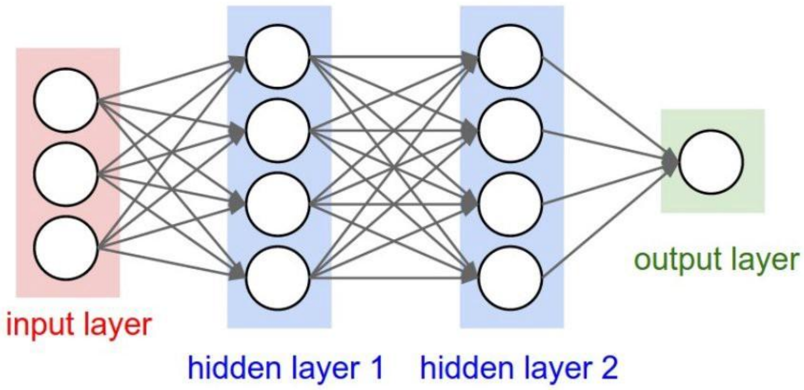
These hot spots are areas where crime is more likely to occur, and law enforcement agencies often focus their resources and efforts in these areas to prevent and reduce criminal activity. Crime hot spots can be identified through various methods such as analysing crime data, conducting community surveys, and mapping criminal activity patterns. By identifying crime hot spots, law enforcement can use targeted strategies to prevent and reduce crime in those areas. The goal of this project is to develop a Crime Hot Spot Prediction System that specifically focuses on women's safety using Machine Learning. The system will be designed to be user-friendly, accessible, and easy to use, with the aim of empowering women to take control of their safety. The Crime Hot Spot Prediction System using Machine Learning will be a valuable tool for law enforcement agencies, policymakers, and community organizations, enabling them to develop targeted strategies for preventing and reducing crime in hot spot areas.

1.2. PROBLEMS DEFINITION

The problem statement of this project is that women traveling in urban areas are increasingly exposed to various forms of crime such as harassment, assault, stalking, kidnapping, and other violent incidents, particularly in high-risk zones. Existing safety solutions and navigation systems primarily focus on shortest distance or fastest routes without considering crime intensity or dynamic risk factors. Traditional crime analysis methods rely heavily on static historical data and basic statistical models, which fail to capture complex spatial and temporal relationships in criminal activities. As a result, crime predictions are often inaccurate, delayed, or not actionable in real time. This lack of predictive and route-aware safety mechanisms leaves women vulnerable and limits the ability of law enforcement agencies to proactively monitor emerging threats. Moreover, current systems do not provide integrated real-time alerts, risk-based route optimization, or predictive hotspot visualization that can assist users in making informed travel decisions. The absence of intelligent crime forecasting tools forces individuals to depend on manual awareness, personal judgment, or outdated crime reports, increasing the likelihood of exposure to unsafe areas. To overcome these challenges, there is a need for a real-time, intelligent, and predictive safety platform capable of accurately identifying emerging crime hotspots, evaluating route-level risk, and providing immediate alerts and safe route recommendations. By leveraging Spatio-Temporal Graph Convolutional Networks (ST-GCN), dynamic geospatial mapping, and automated notification mechanisms, the proposed CrimeAware system offers a scalable and proactive solution. It enhances women’s safety, supports informed decision-making during travel, and assists law enforcement agencies in implementing preventive and data-driven crime control strategies.

1.3. DEEP LEARNING

Deep Learning is transforming the way machines understand, learn and interact with complex data. Deep learning mimics neural networks of the human brain, it enables computers to autonomously uncover patterns and make informed decisions from vast amounts of unstructured data. Neural network consists of layers of interconnected nodes or neurons that collaborate to process input data. In a fully connected deep neural network data flows through multiple layers where each neuron performs nonlinear transformations, allowing the model to learn intricate representations of the data. In a deep neural network, the input layer receives data which passes through hidden layers that transform the data using nonlinear functions. The final output layer generates the model's prediction.

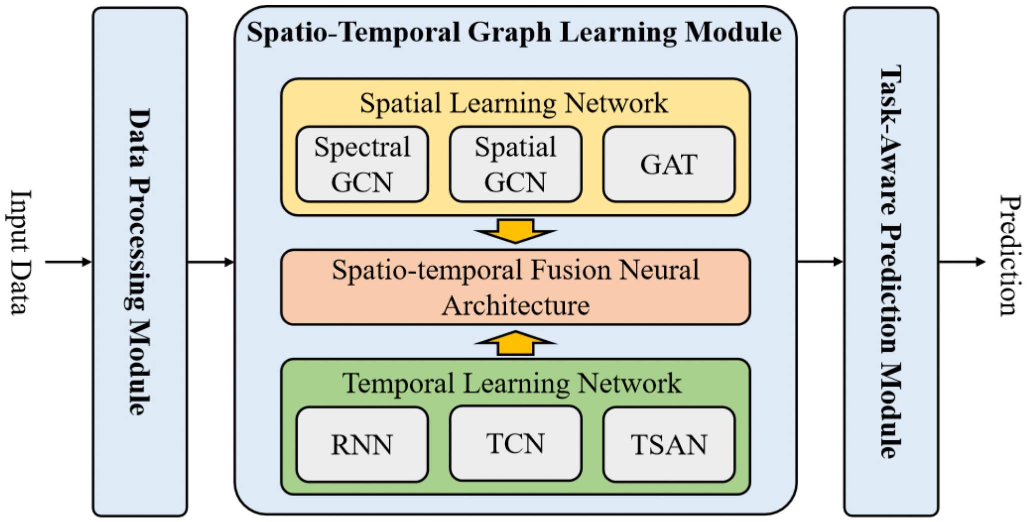


- Feedforward neural networks (FNNs): They are the simplest type of ANN, where data flows in one direction from input to output. It is used for basic tasks like classification.
- Convolutional Neural Networks (CNNs): They are specialized for processing grid-like data, such as images. CNNs use convolutional layers to detect spatial hierarchies, making them ideal for computer vision tasks.
- Recurrent Neural Networks (RNNs): Theyare able to process sequential data, such as time series and natural language. RNNs have loops to retain information over time, enabling applications like language modeling and speech recognition. Variants like LSTMs and GRUs address vanishing gradient issues.
- Generative Adversarial Networks (GANs): This consist of two networks—a generator and a discriminator—that compete to create realistic data. GANs are widely used for image generation, style transfer and data augmentation.
- Autoencoders: They are unsupervised networks that learn efficient data encodings. They compress input data into a latent representation and reconstruct it, useful for dimensionality reduction and anomaly detection.

- Transformer Networks: It has revolutionized NLP with self-attention mechanisms. Transformers excel at tasks like translation, text generation and sentiment analysis, powering models like GPT and BERT.

1.3.3. Spatio-Temporal Graph Convolutional Networks

Spatio-Temporal Graph Convolutional Networks (ST-GCNs) are advanced deep learning models designed to handle data that evolves over time and possesses an inherent spatial structure. They integrate Graph Convolutional Networks (GCNs) to capture spatial dependencies between nodes with Temporal Convolutional Networks (TCNs) or Recurrent Neural Networks (RNNs) like LSTMs to model temporal dynamics. ST-GCNs are particularly effective for applications involving complex, non-stationary patterns, such as traffic forecasting, human action recognition, and skeleton-based motion analysis.



- **Spatial Modeling (GCNs):** GCN layers aggregate information from neighboring nodes within the same time frame, enabling the network to understand structural or relational connections in the data.
- **Temporal Modeling (TCN/RNN):** Temporal layers capture the sequential evolution of each node across multiple time steps, identifying trends and temporal patterns.
- **Architecture:** ST-GCNs are typically built with stacked ST-blocks, where each block contains both spatial and temporal layers, allowing the model to extract joint spatio-temporal features effectively.

1.3.3.1. Applications

- **Traffic Forecasting:** Predicts traffic flow and speed in urban networks by modeling road segments as nodes and their interactions as edges.

- **Human Action Recognition:** Uses skeleton sequences, with joints as nodes and bones as edges, to classify and recognize different human actions.
- **Brain Networks:** Analyzes spatial-temporal features in neuroimaging data to study brain connectivity patterns.
- **Medical Imaging:** Detects key landmarks and temporal changes in sequences like echocardiograms.

1.4. AIM AND OBJECTIVE

Aim

The aim of this project is to develop an intelligent predictive system that identifies high-risk zones for women, provides optimized safe travel routes, and delivers real-time safety alerts using Spatio-Temporal Graph Convolutional Networks (ST-GCN).

Objectives

- To collect and preprocess historical and real-time crime data for women's safety.
- To construct a spatial-temporal graph representing locations and time-based crime patterns.
- To implement the ST-GCN model for predicting high-risk zones using spatial and temporal correlations.
- To classify areas into risk categories (high, medium, low) for effective visualization.
- To perform route analysis and suggest the safest paths from source to destination.
- To provide real-time alerts and notifications to users about potential risks along their route.
- To assist law enforcement agencies in proactive monitoring and decision-making for safer urban environments.

1.5. SCOPE OF THE PROJECT

The project is designed to enhance the safety, awareness, and decision-making capabilities of women traveling in urban environments by predicting crime hotspots and recommending safer routes. The system leverages advanced technologies such as Spatio-Temporal Graph Convolutional Networks (ST-GCN), geospatial data processing, and real-time map visualization to ensure accurate crime risk prediction and proactive safety guidance.

- **Predictive Crime Hotspot Identification:** Uses ST-GCN to analyze historical and temporal crime data to detect high-risk zones and emerging crime patterns with improved accuracy.

- **Risk-Based Route Optimization:** Evaluates multiple travel routes between selected source and destination points and recommends the safest route based on cumulative crime risk instead of just shortest distance.
- **Interactive Map Visualization:** Displays predicted crime hotspots using color-coded heatmaps and spatial overlays, enabling users to clearly understand risk levels across different areas.
- **Real-Time Safety Alerts:** Provides instant notifications when users approach high-risk zones or deviate from recommended safe paths, improving situational awareness.
- **Law Enforcement Monitoring Support:** Offers predictive analytics dashboards to authorities for hotspot trend analysis, patrolling strategy planning, and proactive crime prevention.
- **Secure Web Application Interface:** Ensures authenticated access for administrators, general users, and law enforcement agencies with efficient backend integration and dynamic content rendering.
- **Scalability and Future Expansion:** Designed to support integration with mobile applications, emergency response systems, surveillance infrastructure, and expansion to additional cities or crime categories.

This project aims to transform traditional navigation systems into intelligent safety-oriented platforms, strengthen preventive crime monitoring, and contribute to building safer urban environments for women.

CHAPTER 2
SYSTEM ANALYSIS

2.1. EXISTING SYSTEM

There are several existing systems for Crime Hot Spot Prediction, but very few systems specifically focus on predicting crimes against women. Some of the existing systems are:

- **Traditional Crime Analysis**

Most existing systems depend on historical crime records and simple statistical models, which offer limited ability to capture changing crime patterns. These methods do not account for spatial or temporal variations, resulting in outdated and less accurate hotspot identification.

- **Navigation and Mapping Apps**

Current navigation platforms focus mainly on shortest distance or fastest travel time. They do not evaluate route safety or consider crime-prone areas, leaving users—especially women—unaware of potential risks along the path they choose.

- **Safety Applications**

Many safety apps provide incident reporting or panic alert features but lack predictive capabilities. They notify users only after a crime occurs, offering minimal support in preventing them from entering high-risk zones beforehand.

- **Law Enforcement Monitoring**

Law enforcement agencies often rely on manual hotspot mapping and periodic reviews of crime data. This process is slow and reactive, making it difficult to detect emerging crime trends or respond proactively to rising risks in specific locations.

2.1.1. Disadvantages

- Existing systems provide low prediction accuracy due to simple statistical models.
- Lack of real-time crime updates leads to delayed hotspot detection.
- Navigation apps do not evaluate route safety or highlight crime-prone areas.
- Safety tools are reactive, issuing alerts only after incidents occur.
- Law enforcement lacks advanced tools for monitoring evolving crime patterns.

2.2. PROPOSED SYSTEM

The proposed system introduces a predictive, real-time safety platform designed to enhance women's security during travel by identifying crime hotspots, evaluating route risks, and providing timely safety alerts.

- **Predictive Hotspot Detection**

The system employs Spatio-Temporal Graph Convolutional Networks (ST-GCN) to analyze both spatial and temporal crime patterns, enabling precise identification of emerging high-risk zones in real time. By incorporating historical and live incident data, the model ensures dynamic and adaptive hotspot prediction.

- **Safe Route Optimization**

Instead of focusing solely on distance or travel time, multiple routes are evaluated based on predicted risk levels. The system recommends the safest path, reducing exposure to crime-prone areas while still considering travel efficiency and user convenience.

- **Real-Time Safety Alerts**

Users receive instant notifications when approaching high-risk areas or when recent incidents occur along their route. These alerts enhance situational awareness and allow users to make informed decisions during travel, increasing overall personal safety.

- **Proactive Law Enforcement Support**

Authorities can access predicted hotspot trends and receive insights into potential crime escalation, enabling efficient allocation of resources, preemptive patrolling, and faster response times, thereby strengthening urban safety management.

2.2.1. Advantages

- Accurately predicts crime hotspots using spatio-temporal analysis.
- Suggests safest routes by evaluating risk levels.
- Provides real-time alerts for nearby incidents.
- Updates risk maps dynamically with live data.
- Helps law enforcement with early hotspot detection.
- Easy map-based visualization for users.
- Enhances overall public safety through integrated prediction and alerts.

CHAPTER 3
SYSTEM STUDY

3.1. INTRODUCTION

The safety of women in urban environments has become a major social concern due to the increasing number of crimes such as harassment, assault, kidnapping, stalking, domestic violence, and trafficking occurring in public spaces and high-risk locations. Traditional safety systems and crime prediction methods mainly depend on historical records and static statistical analysis, which are often unable to accurately predict rapidly changing crime patterns or identify emerging danger zones in real time. To overcome these challenges, this project proposes an intelligent Women Safety Crime Hotspot Prediction and Safe Route Recommendation System using Spatio-Temporal Graph Convolutional Networks (ST-GCN). The system analyzes both spatial and temporal relationships in crime data to dynamically identify crime-prone hotspots and predict potential risks across different urban regions. By integrating historical crime records with live incident updates, the platform continuously monitors changing crime trends and generates real-time hotspot maps. When a user enters a source and destination, the system evaluates multiple travel routes, detects unsafe areas along each path, and recommends the safest route with minimum risk exposure while also providing estimated travel time. The platform further enhances security by issuing alerts regarding recent criminal activities near the selected route and notifying local authorities if abnormal or high-risk situations are detected. In addition, the system can include emergency SOS support, live location sharing, and risk-level visualization using interactive maps, allowing users to stay informed throughout their journey. The proposed framework can also help police departments and city administrators identify vulnerable zones, optimize patrol planning, and take preventive measures in crime-prone regions. Through advanced deep learning, real-time analytics, and intelligent route planning, the proposed system improves women's personal safety, supports proactive law enforcement, increases public risk awareness, and contributes to building safer and smarter cities while promoting confidence and secure mobility for women in public spaces.

3.2 FEASIBILITY STUDY

The feasibility study evaluates the practicality and effectiveness of the proposed Women Safety Crime Hotspot Prediction System. The system uses ST-GCN, GPS navigation, and real-time crime analytics to provide safe route recommendations and hotspot detection. It is technically, economically, and operationally feasible for deployment in smart city and women safety applications.

3.2.1 TECHNICAL FEASIBILITY

The proposed system is technically feasible as it uses existing technologies such as Python, deep learning frameworks, GIS mapping tools, GPS navigation, cloud databases, and real-time communication services. The ST-GCN model effectively analyzes spatial and temporal crime patterns to predict crime hotspots accurately. Historical and live crime data can be integrated through APIs and cloud platforms for continuous monitoring and route safety analysis. Mapping services like Google Maps or OpenStreetMap support safe route visualization, while modern mobile devices can efficiently handle notifications, alerts, and live tracking features. Therefore, the system can be successfully implemented and scaled for urban environments.

3.2.2 ECONOMIC FEASIBILITY

The system is economically feasible because it mainly depends on software technologies, open-source tools, and publicly available crime datasets, reducing development and infrastructure costs. Existing smartphones, internet services, and GPS devices can support the platform without requiring expensive hardware. The system also helps reduce costs related to crime prevention and emergency response by improving resource management and patrol planning. Hence, the project is affordable and practical for both government and private organizations.

3.2.3 OPERATIONAL FEASIBILITY

The proposed system is operationally feasible as it is designed to be simple, reliable, and user-friendly for both women users and law enforcement agencies. Users can easily access safe route recommendations, hotspot alerts, and emergency support features through an interactive interface. The system continuously updates crime information and dynamically adjusts safety recommendations in real time. Authorities can also use the platform for monitoring crime-prone areas and taking preventive actions, making the system effective for real-world deployment in smart city and public safety applications.

CHAPTER 4
SYSTEM REQUIREMENTS

4.1. HARDWARE REQUIREMENTS

- **Processor** : Intel i5 or higher
- **RAM** : 8 GB minimum
- **Storage** : 500 GB HDD / SSD
- **GPU** : NVIDIA GPU (for deep learning model training, optional for deployment)
- **Internet Connection** : Stable broadband for real-time data fetching
- **Display** : 15-inch monitor or higher

4.2. SOFTWARE REQUIREMENTS

- **Operating System** : Windows 10 / 11
- **Programming** : Python 3.9+
- **Web Framework** : Flask
- **Database** : MySQL
- **Frontend** : HTML, CSS, Bootstrap
- **Libraries** : Pandas, NumPy, Geopandas, Folium, Matplotlib, Scikit-learn, TensorFlow/PyTorch
- **Tools** : Google Maps API for visualization and route plotting

4.3. SOFTWARE DESCRIPTION

1.PYTHON 3.7.4

Python is a general-purpose interpreted, interactive, object-oriented, and high-level programming language. It was created by Guido van Rossum during 1985- 1990. Like Perl, Python source code is also available under the GNU General Public License (GPL). This tutorial gives enough understanding on Python programming language.

Python is a high-level, interpreted, interactive and object-oriented scripting language. Python is designed to be highly readable. It uses English keywords frequently where as other languages use punctuation, and it has fewer syntactical constructions than other languages. Python is a MUST for students and working professionals to become a great Software Engineer specially when they are working in Web Development Domain.

Python is currently the most widely used multi-purpose, high-level programming language. Python allows programming in Object-Oriented and Procedural paradigms. Python programs generally are smaller than other programming languages like Java. Programmers have to type relatively less and indentation requirement of the language, makes them readable all the time. Python language is being used by almost all tech-giant companies like – Google, Amazon, Facebook, Instagram, Dropbox, Uber… etc. The biggest strength of Python is huge collection of standard libraries which can be used for the following:

- Machine Learning
- GUI Applications (like Kivy, Tkinter, PyQt etc.)
- Web frameworks like Django (used by YouTube, Instagram, Dropbox)
- Image processing (like OpenCV, Pillow)
- Web scraping (like Scrapy, BeautifulSoup, Selenium)
- Test frameworks
- Multimedia
- Scientific computing, Text processing and many more.

Tensor Flow

TensorFlow is an end-to-end open-source platform for machine learning. It has a comprehensive, flexible ecosystem of tools, libraries, and community resources that lets researchers push the state-of-the-art in ML, and gives developers the ability to easily build and deploy ML-powered applications.

TensorFlow provides a collection of workflows with intuitive, high-level APIs for both beginners and experts to create machine learning models in numerous languages. Developers have the option to deploy models on a number of platforms such as on servers, in the cloud, on mobile and edge devices, in browsers, and on many other JavaScript platforms. This enables developers to go from model building and training to deployment much more easily.

Pandas

pandas are a fast, powerful, flexible and easy to use open source data analysis and manipulation tool, built on top of the Python programming language. pandas are a Python package that provides fast, flexible, and expressive data structures designed to make working with "relational" or "labeled" data both easy and intuitive. It aims to be the fundamental high-level building block for doing practical, real world data analysis in Python.

Pandas is mainly used for data analysis and associated manipulation of tabular data in Data frames. Pandas allows importing data from various file formats such as comma-separated values, JSON, Parquet, SQL database tables or queries, and Microsoft Excel. Pandas allows various data manipulation operations such as merging, reshaping, selecting, as well as data cleaning, and data wrangling features. The development of pandas introduced into Python many comparable features of working with Data frames that were established in the R programming language. The panda’s library is built upon another library NumPy, which is oriented to efficiently working with arrays instead of the features of working on Data frames.

NumPy

NumPy, which stands for Numerical Python, is a library consisting of multidimensional array objects and a collection of routines for processing those arrays. Using NumPy, mathematical and logical operations on arrays can be performed.

NumPy is a general-purpose array-processing package. It provides a high-performance multidimensional array object, and tools for working with these arrays.

Matplotlib

Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python. Matplotlib makes easy things easy and hard things possible.

Matplotlib is a plotting library for the Python programming language and its numerical mathematics extension NumPy. It provides an object-oriented API for embedding plots into applications using general-purpose GUI toolkits like Tkinter, wxPython, Qt, or GTK.

Scikit Learn

scikit-learn is a Python module for machine learning built on top of SciPy and is distributed under the 3-Clause BSD license.

Scikit-learn (formerly scikits. learn and also known as sklearn) is a free software machine learning library for the Python programming language. It features various classification, regression and clustering algorithms including support-vector machines, random forests, gradient boosting, k-means and DBSCAN, and is designed to interoperate with the Python numerical and scientific libraries NumPy and SciPy.

2.MYSQL

MySQL is a relational database management system based on the Structured Query Language, which is the popular language for accessing and managing the records in the database. MySQL is open-source and free software under the GNU license. It is supported by Oracle Company. MySQL database that provides for how to manage database and to manipulate data with the help of various SQL queries. These queries are: insert records, update records, delete records, select records, create tables, drop tables, etc. There are also given MySQL interview questions to help you better understand the MySQL database.

MySQL is currently the most popular database management system software used for managing the relational database. It is open-source database software, which is supported by Oracle Company. It is fast, scalable, and easy to use database management system in comparison with Microsoft SQL Server and Oracle Database. It is commonly used in conjunction with PHP scripts for creating powerful and dynamic server-side or web-based enterprise applications. It is developed, marketed, and supported by MySQL AB, a Swedish company, and written in C programming language and C++ programming language. The official pronunciation of MySQL is not the My Sequel; it is My Ess Que Ell. However, you can pronounce it in your way. Many small and big companies use MySQL. MySQL supports many Operating Systems like Windows, Linux, MacOS, etc. with C, C++, and Java languages.

3.WAMPSERVER

WampServer is a Windows web development environment. It allows you to create web applications with Apache2, PHP and a MySQL database. Alongside, PhpMyAdmin allows you to manage easily your database.

WAMPServer is a reliable web development software program that lets you create web apps with MYSQL database and PHP Apache2. With an intuitive interface, the application features numerous functionalities and makes it the preferred choice of developers from around the world. The software is free to use and doesn't require a payment or subscription.

4.BOOTSTRAP 4

Bootstrap is a free and open-source tool collection for creating responsive websites and web applications. It is the most popular HTML, CSS, and JavaScript framework for developing responsive, mobile-first websites.

It solves many problems which we had once, one of which is the cross-browser compatibility issue. Nowadays, the websites are perfect for all the browsers (IE, Firefox, and Chrome) and for all sizes of screens (Desktop, Tablets, Phablets, and Phones). All thanks to Bootstrap developers -Mark Otto and Jacob Thornton of Twitter, though it was later declared to be an open-source project.

Easy to use: Anybody with just basic knowledge of HTML and CSS can start using Bootstrap
Responsive features: Bootstrap's responsive CSS adjusts to phones, tablets, and desktops
Mobile-first approach: In Bootstrap, mobile-first styles are part of the core framework
Browser compatibility: Bootstrap 4 is compatible with all modern browsers (Chrome, Firefox, Internet Explorer 10+, Edge, Safari, and Opera)

5.FLASK

Flask is a web framework. This means flask provides you with tools, libraries and technologies that allow you to build a web application. This web application can be some web pages, a blog, a wiki or go as big as a web-based calendar application or a commercial website.

Flask is often referred to as a micro framework. It aims to keep the core of an application simple yet extensible. Flask does not have built-in abstraction layer for database handling, nor does it have formed a validation support. Instead, Flask supports the extensions to add such functionality to the application. Although Flask is rather young compared to most Python frameworks, it holds a great promise and has already gained popularity among Python web developers. Let’s take a closer look into Flask, so-called “micro” framework for Python. Flask is part of the categories of the micro-framework. Micro-framework are normally framework with little to no dependencies to external libraries. This has pros and cons. Pros would be that the framework is light, there are little dependency to update and watch for security bugs, cons is that some time you will have to do more work by yourself or increase yourself the list of dependencies by adding plugins.

CHAPTER 5
SYSTEM DESIGN

5.1. PROJECT DESCRIPTION

This project proposes an intelligent women's safety system that predicts and maps crime hotspots using advanced Spatio-Temporal Graph Convolutional Networks (ST-GCN), enabling real-time analysis of both historical and live crime data to identify high-risk areas with greater accuracy than traditional methods. The system allows users to input their source and destination, evaluates multiple travel routes, and highlights paths that pass through crime-prone zones while recommending safer alternatives along with estimated travel time. It continuously updates risk levels based on recent incidents such as harassment, assault, kidnapping, or suspicious activities, ensuring dynamic and context-aware guidance, while integrating GPS tracking, mobile connectivity, and cloud-based processing for seamless real-time monitoring. The platform also features an emergency alert mechanism that allows users to share their live location with trusted contacts and authorities during critical situations, and it can automatically notify law enforcement when unusual crime patterns are detected for faster preventive action. Visualization tools like heat maps and risk indicators help users easily understand unsafe zones, while crowd-sourced inputs and user feedback further enhance prediction accuracy. In addition, the system can incorporate time-based risk analysis, identifying patterns such as higher crime probability during late-night hours or in poorly lit areas, and can suggest precautionary measures before travel. The model is scalable across multiple cities and supports integration with smart city infrastructure such as surveillance systems and public safety networks. Continuous learning capabilities ensure that the system adapts over time, improving prediction performance as more data is collected. By combining predictive analytics, geospatial intelligence, route optimization, and real-time alerts, the system enhances personal safety, supports proactive policing, and empowers women to make informed and confident travel decisions, ultimately contributing to safer, smarter, and more secure urban environments.

5.2. MODULES EXPLANATION

1. CrimeAware Web App

This module is responsible for the complete development and deployment of the CrimeAware web application. It is built using Python, Flask, MySQL, Bootstrap, and WAMP Server, ensuring a scalable and responsive system. The module establishes communication between the user interface and backend processes, enabling seamless access to prediction services, route analysis, and map visualization. It manages user sessions, processes input data, handles backend queries, and renders visual outputs such as hotspot maps and recommended routes. This module also integrates APIs (such as Google Maps) and necessary Python libraries for geospatial analysis. It ensures secure login mechanisms, efficient database operations, and dynamic page rendering. It forms the foundation of the system, enabling end users, administrators, and law enforcement agencies to interact with the platform in real time.

2. End User

2.1 Server Admin

This module allows the system administrator to oversee and manage the entire application. The admin can log in securely and upload new crime datasets, ensuring the model is trained on the latest information. The admin is responsible for triggering model training processes, monitoring training progress, and validating model accuracy. Additionally, the Admin manages user accounts, updates permissions, and ensures proper access control for both general users and law enforcement agencies. The module also includes tools for monitoring system performance, reviewing logs, updating backend settings, and maintaining data consistency. This ensures that the CrimeAware system remains up-to-date, secure, and functioning optimally.

2.2 Women User / General User

This module enables women or general users to interact with the system after registering and logging in. They can input a single desired location to view crime hotspots in that region or select both source and destination to receive route-based safety guidance. The module communicates with backend prediction services to display risk zones on interactive maps. Users receive route recommendations based on risk analysis and are alerted instantly when deviating into unsafe areas.

The module ensures user-friendly interaction by presenting clear map visualizations, risk indicators, and step-by-step route suggestions. It is designed to enhance awareness, safety decision-making, and navigation confidence.

2.3 Law Enforcement Agency

This module allows authorized law enforcement personnel to log in and access predictive analytics and hotspot trends. They can review spatial distributions of high-risk zones, analyze historical and predicted crime statistics, and monitor real-time alert reports from users. This enables authorities to proactively identify potential danger zones, plan patrolling strategies, allocate resources, and respond quickly to emerging threats. The module acts as a bridge between public safety applications and operational law enforcement workflows by providing actionable insights and visual hotspot intelligence.

3. Crime Hot Spot Analyzer

This module focuses on training a classification model using Spatio-Temporal Graph Convolutional Networks (ST-GCN) to categorize regions based on crime risk levels.

3.1 Dataset Annotation

Raw crime data often contains inconsistencies and missing values. The annotation step involves labeling each entry with accurate timestamps, geographical coordinates, crime types, and severity indicators. The dataset is segmented into time intervals (hour/day/week) and spatial blocks (regions/grid cells) for proper graph construction. This ensures uniformity and structured representation of the data for ST-GCN.

3.2 Preprocessing

In this step, noise and redundant data are removed. Spatial clustering is applied to group nearby crime events, and temporal ordering is enforced to capture sequential patterns. Missing values are filled using interpolation techniques. The module prepares adjacency matrices representing spatial relationships and time-series matrices representing temporal progression. Preprocessing ensures data quality and compatibility with the graph-based neural network.

3.3 Feature Selection & Extraction

This component identifies key crime-related features such as frequency trends, seasonal patterns, geographic clustering, time-of-day variations, and type-specific risk indicators. Features are extracted to capture spatial dependencies and temporal behaviors effectively.

The resulting spatio-temporal feature vectors are used to construct graph nodes, enabling the ST-GCN model to learn meaningful patterns.

3.4 Classification

Using the extracted features, the ST-GCN classifier analyzes crime patterns and assigns risk categories (e.g., low, moderate, high) to different locations. It learns correlations between connected regions and detects repeated or emerging crime behaviors over time. The classifier outputs labeled risk zones that form the basis for prediction and map visualization.

3.5 Train and Build Model

The processed dataset is used to train the ST-GCN model iteratively. The model is optimized using validation metrics, reducing prediction errors and improving classification accuracy. Once training is complete, the model is saved and integrated into the main system. It becomes capable of running classification tasks in real time, supporting hotspot prediction and route analysis.

4. Crime Hot Spot Predictor

This module extends the classifier to perform real-time prediction of potential crime hotspots.

4.1 Input Location or Route Selection

Users or law enforcement personnel can input a specific location, choose coordinates, or select source–destination points on the map. This input triggers prediction queries handled by the ST-GCN model, which evaluates crime risk around the selected areas.

4.2 Hot Spot Prediction

The ST-GCN model predicts crime intensity for different time windows and spatial regions. It analyzes ongoing trends, temporal sequences, and neighborhood influences to generate accurate hotspot forecasts. The output is a risk probability map highlighting potential danger zones. Predictions are used for both visualization and route safety scoring.

5. Crime Hot Spot Map Visualization

This module translates prediction data into interactive geospatial visualizations. It uses mapping libraries to display hotspots as color-coded heatmaps, markers, and dynamic overlays. Users can zoom, pan, and explore risk levels across the city. The visualization clearly differentiates safe, moderate, and high-risk regions.

It improves user understanding, supports immediate decision-making, and helps law enforcement plan interventions. Visual cues, legends, intensity gradients, and layer controls make hotspot representation intuitive and easy to interpret.

6. Route Analysis & Safe Path

This module evaluates all possible travel routes between the user's chosen points. Using risk scores from the predictor, each route is assessed based on cumulative crime intensity. The module identifies hotspot segments, computes a safety index for every route, and recommends the safest path instead of the shortest one. It highlights dangerous areas along the route and provides alternative paths that minimize exposure to crime-prone locations. This intelligent routing system transforms traditional navigation into a safety-oriented decision-making tool for women.

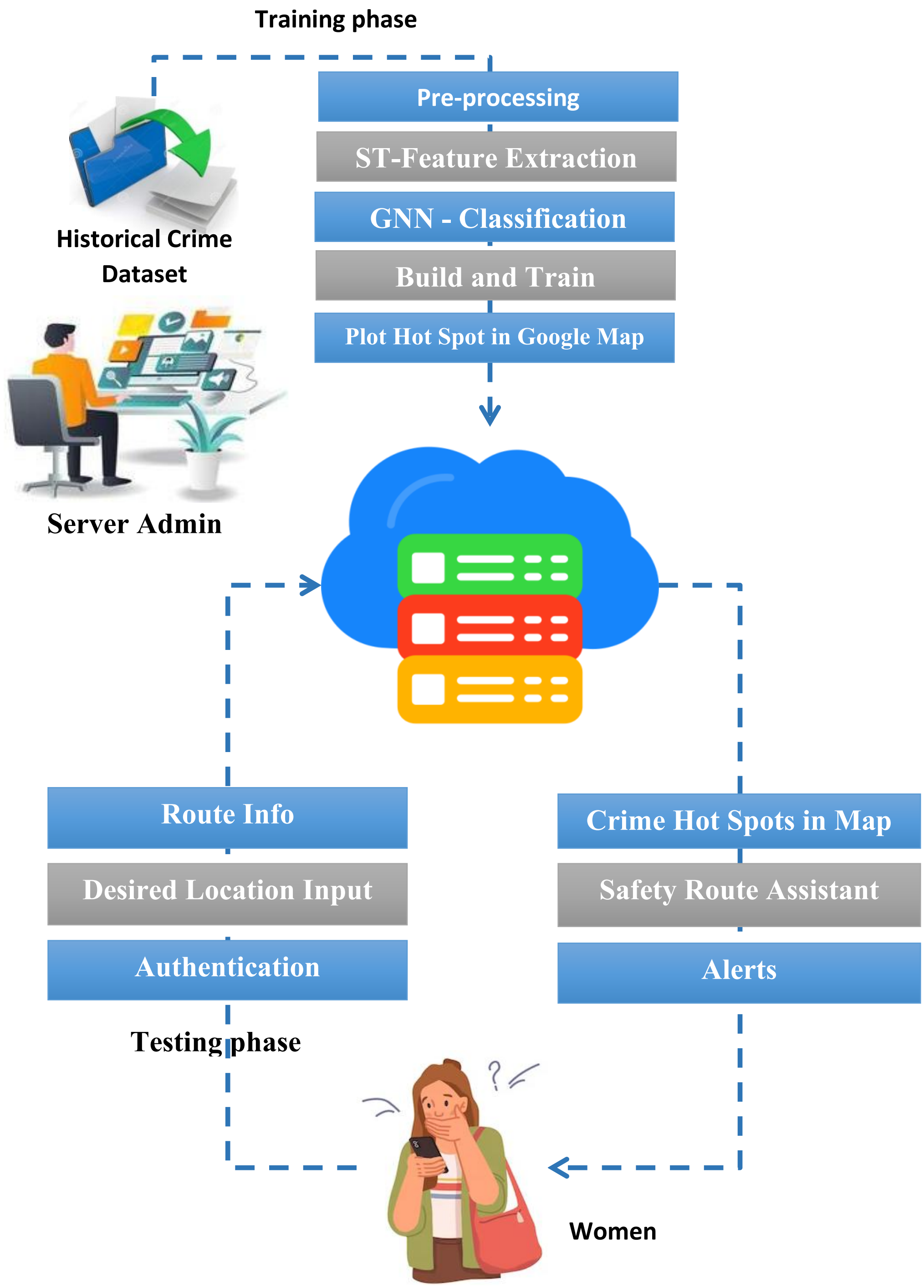
7. Alert & Notification

This module is designed for real-time safety communication. Users receive immediate alerts when approaching predicted hotspots or deviating from safe routes. Notifications include recent incident details, hazard levels, and recommended actions. For law enforcement, the module generates periodic hotspot alerts, emerging risk trend reports, and deviation-based signals for unusual movements or patterns. This ensures timely awareness, proactive responses, and enhanced safety.

CHAPTER 6

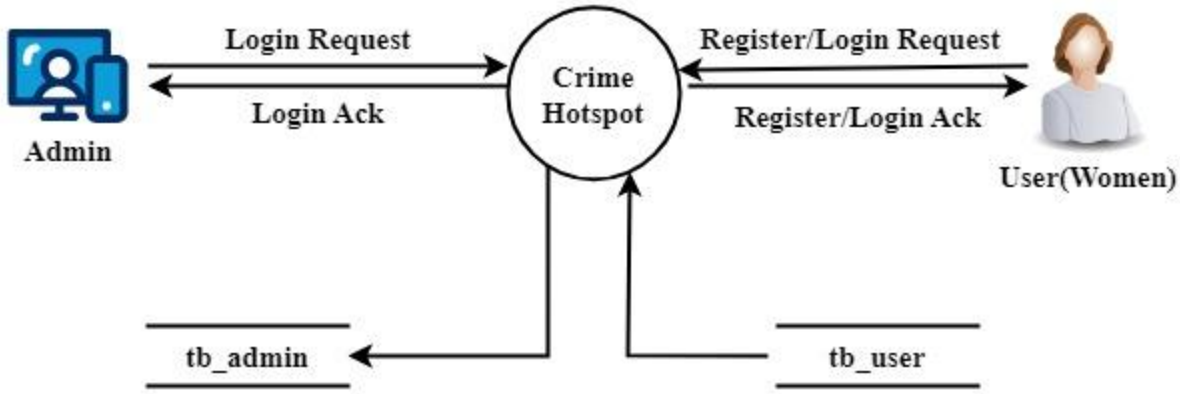
SYSTEM MODELING

6.1. SYSTEM ARCHITECTURE

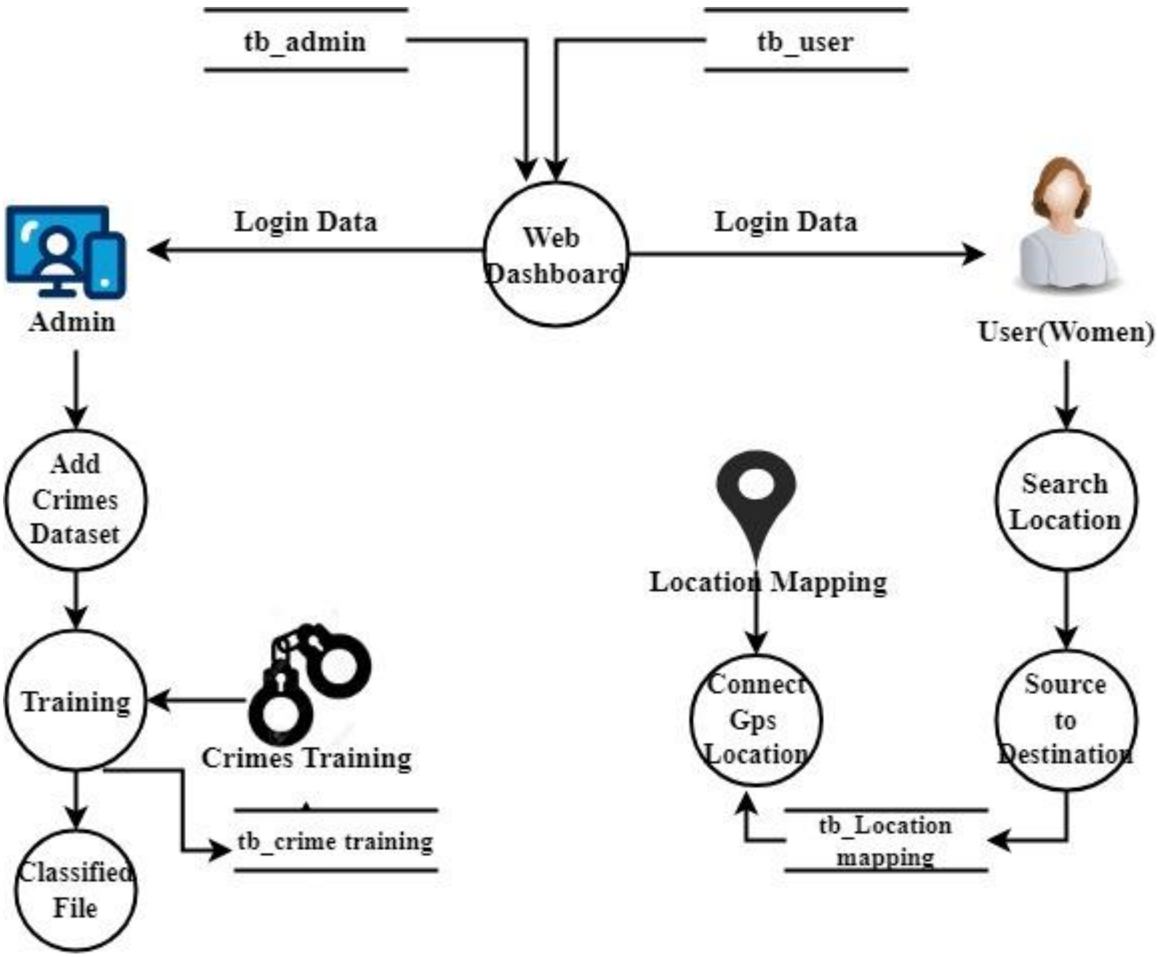


6.2. DATA FLOW DIAGRAM

LEVEL 0

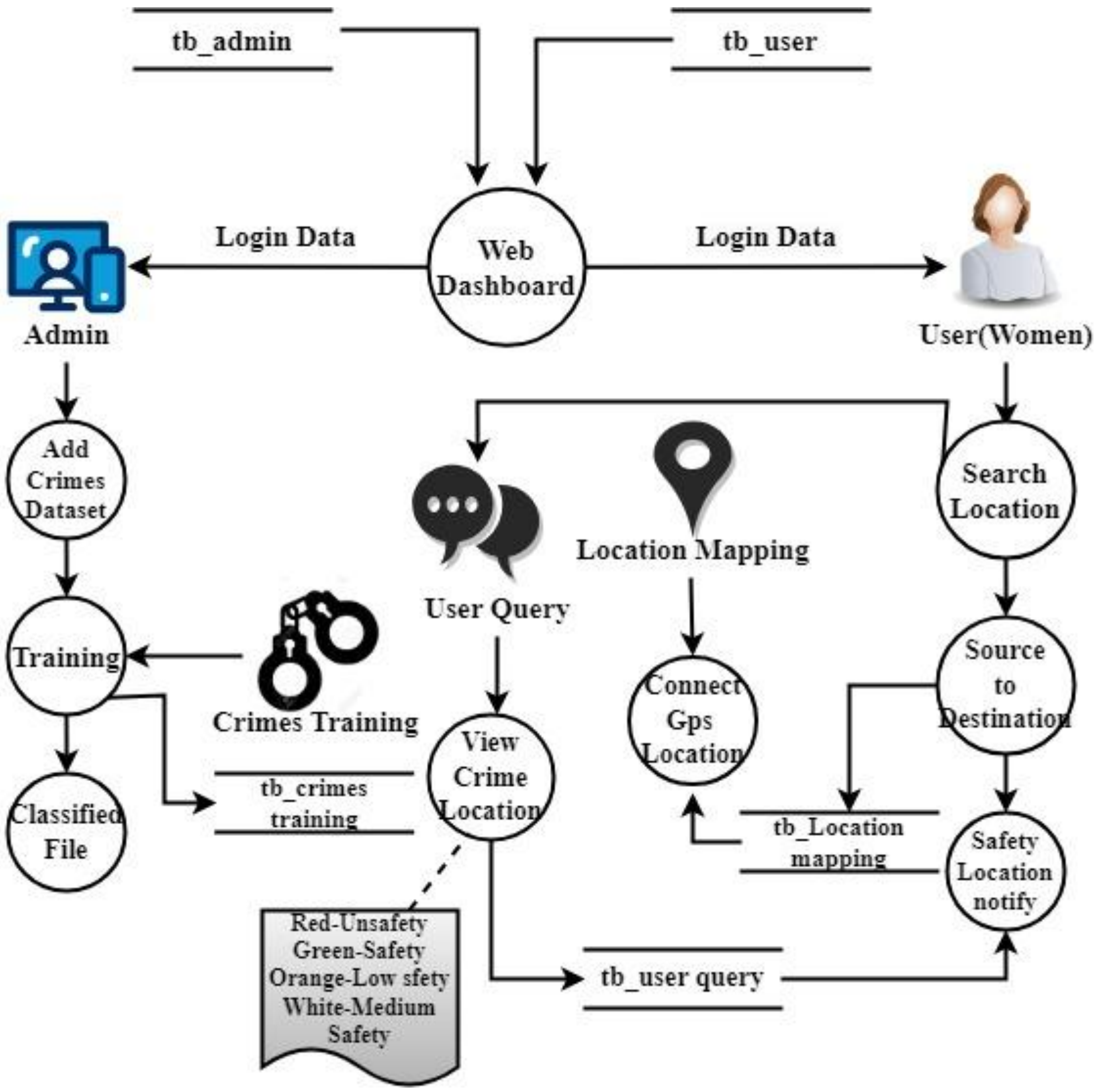


LEVEL 1



tb_admin
tb_user
Login Data
Web Dashboard
Login Data
Admin
User(Women)
Add Crimes Dataset
Search Location
Location Mapping
Training
Crimes Training
Connect Gps Location
Source to Destination
tb_crime training
tb_Location mapping
Classified File

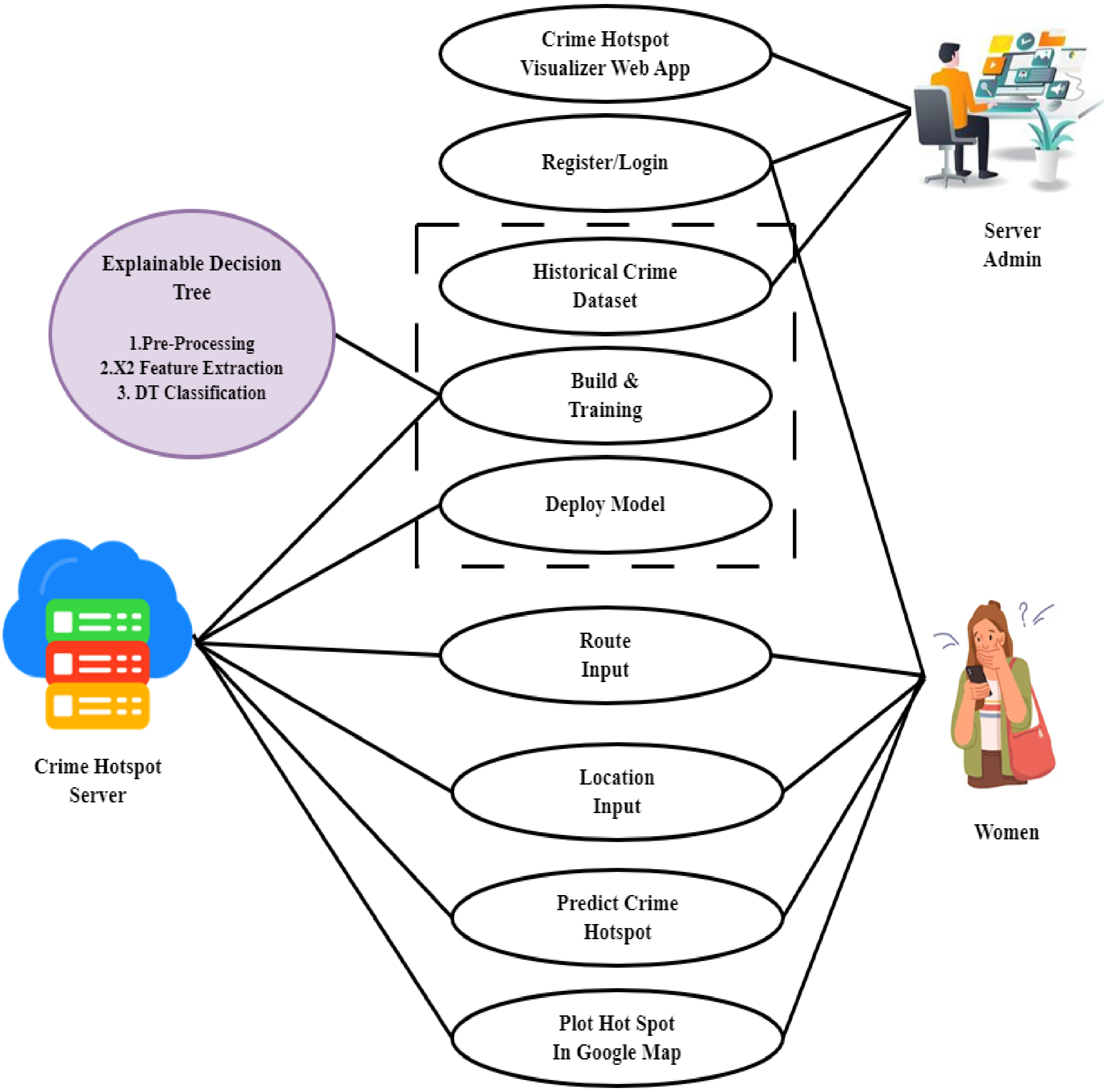
LEVEL 2



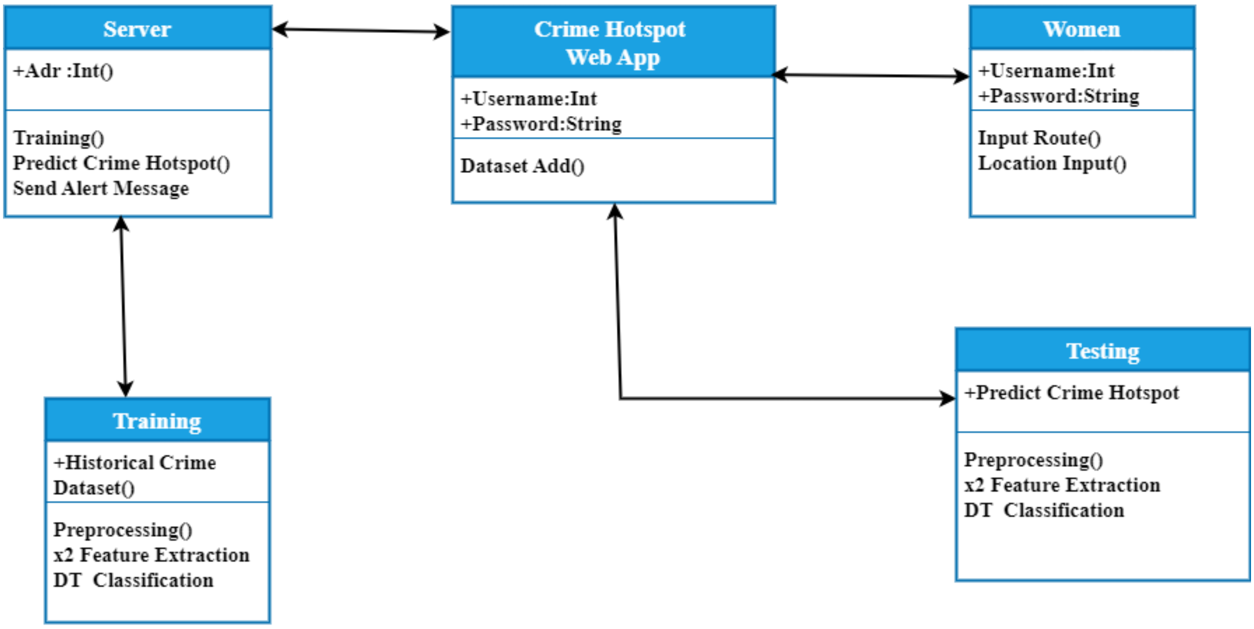
tb_admin
tb_user
Login Data
Web Dashboard
Login Data
Admin
User(Women)
Add Crimes Dataset
Search Location
User Query
Location Mapping
Training
Crimes Training
Connect Gps Location
Source to Destination
View Crime Location
Classified File
tb_crimes training
tb_Location mapping
Safety Location notify
Red-Unsafety
Green-Safety
Orange-Low sfety
White-Medium Safety
tb_user query

6.3.UML DIAGRAM

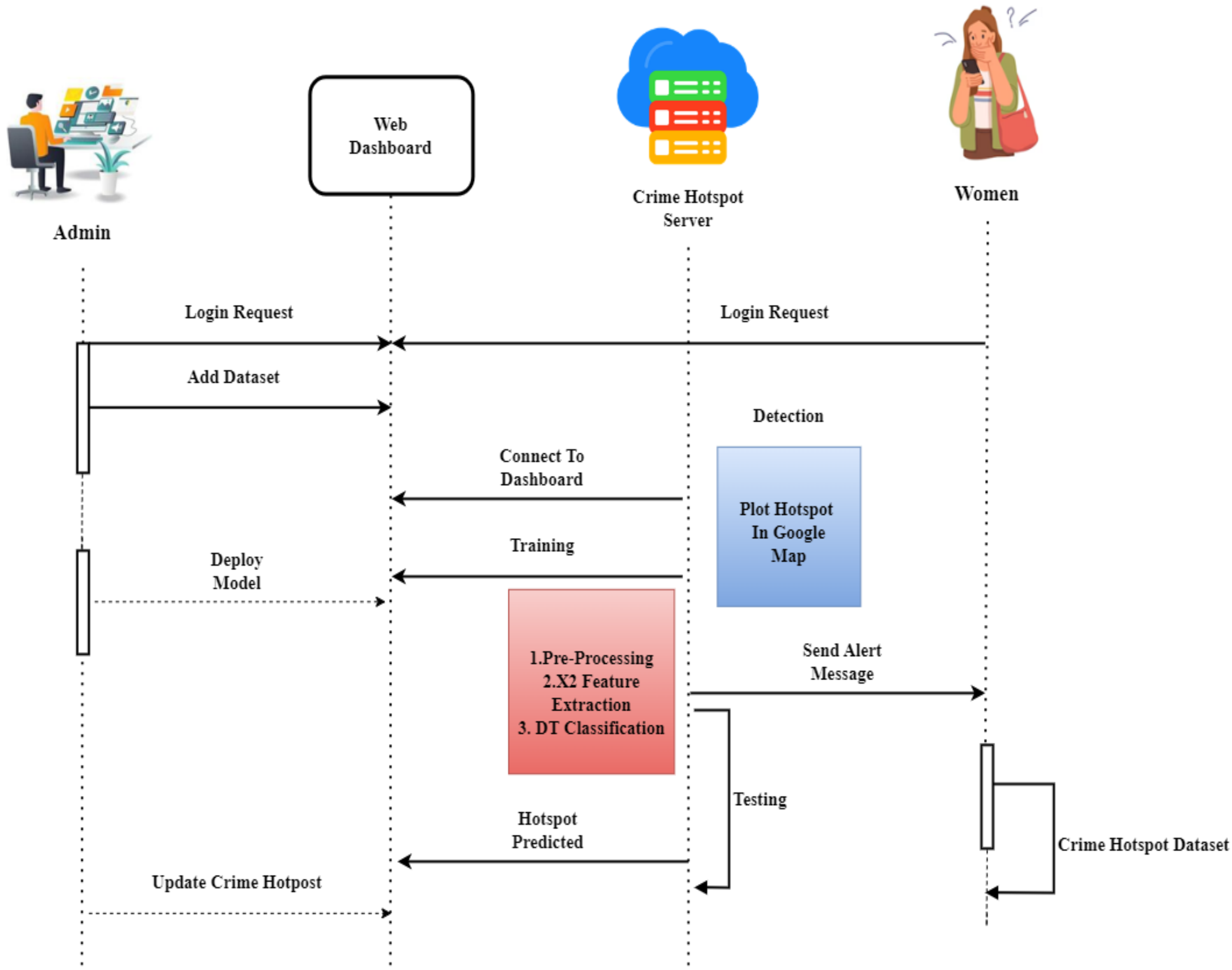
6.3.1.USE CASE DIAGRAM



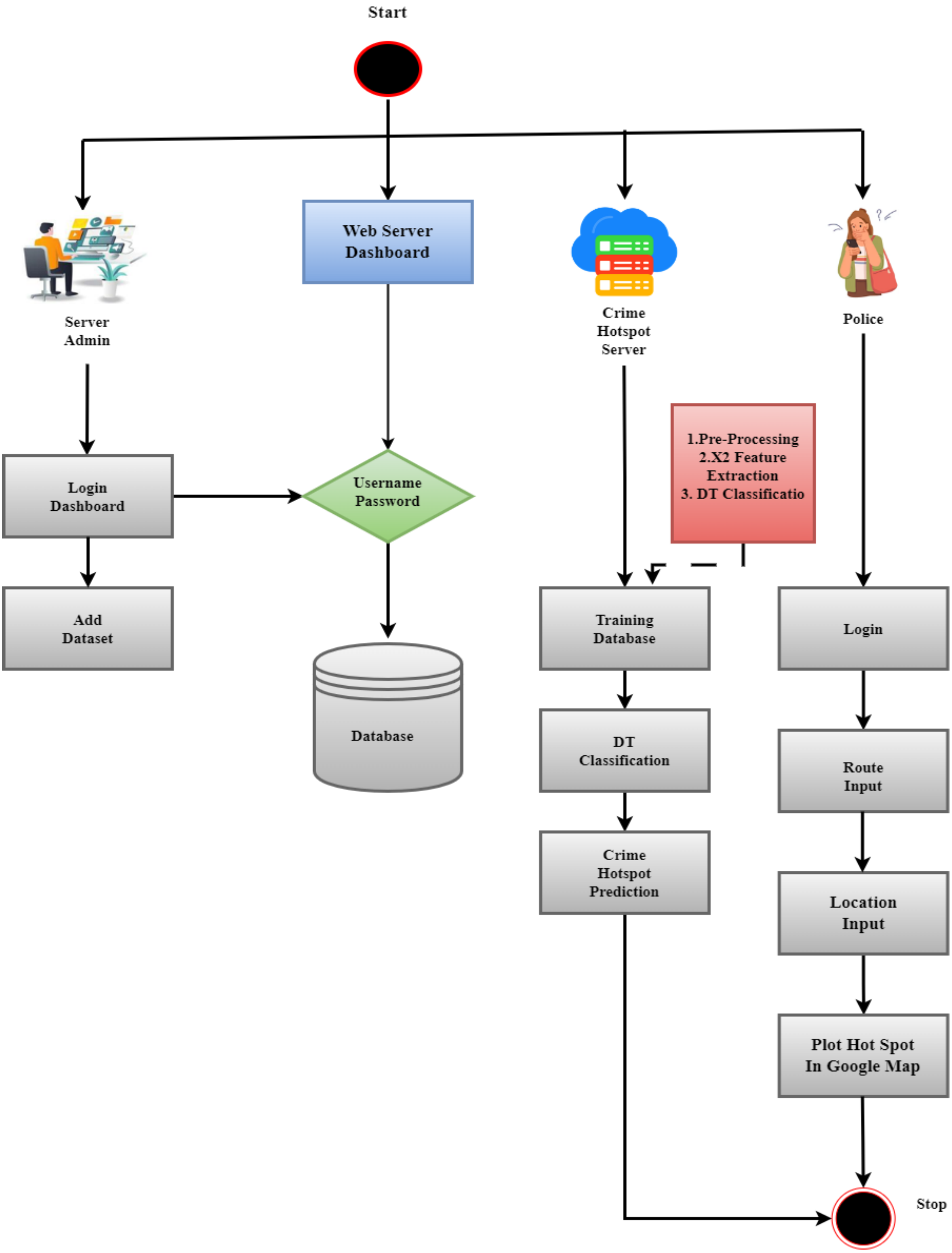
6.3.2.CLASS DIAGRAM



6.3.3. SEQUENCE DIAGRAM


Admin
Web Dashboard
Crime Hotspot Server
Women
Login Request
Login Request
Add Dataset
Detection
Connect To Dashboard
Plot Hotspot In Google Map
Training
Deploy Model
1.Pre-Processing
2.X2 Feature Extraction
3. DT Classification
Send Alert Message
Testing
Crime Hotspot Dataset
Hotspot Predicted
Update Crime Hotpost

6.3.4. ACTIVITY DIAGRAM



6.4. DATA BASE DESIGN

Admin Table

Field Name	Data Type (Size, Constraints)	Description
Admin Id	Integer (Primary Key, Not Null)	Unique Admin Identifier
Full Name	Varchar (100, Not Null)	Admin Name
Email Id	Varchar (100, Unique, Not Null)	Admin Email
Password	Varchar (255, Not Null)	Admin Password
Phone Number	Varchar (20)	Contact Number
Created Date	Datetime (Not Null)	Account Creation Time

Women/User Table

Field Name	Data Type (Size, Constraints)	Description
User Id	Integer (Primary Key, Not Null)	Unique User Identifier
Full Name	Varchar (100, Not Null)	User Name
Email Id	Varchar (100, Unique, Not Null)	User Email
Password	Varchar (255, Not Null)	User Password
Phone Number	Varchar (20)	Contact Number
Created Date	Datetime (Not Null)	Registration Time

Law Enforcement Agency Table

Field Name	Data Type (Size, Constraints)	Description
Agency Id	Integer (Primary Key, Not Null)	Unique Agency Identifier
Agency Name	Varchar (150, Not Null)	Department Name
Officer Name	Varchar (100, Not Null)	Assigned Officer
Email Id	Varchar (100, Unique, Not Null)	Agency Email
Password	Varchar (255, Not Null)	Login Password
Phone Number	Varchar (20)	Contact Number
Access Level	Varchar (20, Not Null)	Read / Monitor
Created Date	Datetime (Not Null)	Account Creation Time

Crime Data Table

Field Name	Data Type (Size, Constraints)	Description
Crime Id	Integer (Primary Key, Not Null)	Unique Crime Record
Crime Type	Varchar (50, Not Null)	Category Of Crime
Crime Details	Varchar (255)	Short Description
Latitude	Decimal (10,6, Not Null)	Location Lat
Longitude	Decimal (10,6, Not Null)	Location Long
Crime Date Time	Datetime (Not Null)	Incident Timestamp
Severity Level	Varchar (20)	Risk Level

Model Training Logs Table

Field Name	Data Type (Size, Constraints)	Description
Train Id	Integer (Primary Key, Not Null)	Training Identifier
Dataset Id	Integer (Not Null)	Dataset Reference
Accuracy	Decimal (5,2)	Model Accuracy
Loss Value	Decimal (8,4)	Model Loss
Train Date	Datetime (Not Null)	Training Time
Trained By	Integer (Not Null)	Admin Id Reference

Hotspot Predictions Table

Field Name	Data Type (Size, Constraints)	Description
Prediction Id	Integer (Primary Key, Not Null)	Prediction Identifier
Region Code	Varchar (50, Not Null)	Area Code
Latitude	Decimal (10,6, Not Null)	Predicted Lat
Longitude	Decimal (10,6, Not Null)	Predicted Long
Risk Score	Decimal (4,3, Not Null)	Risk Level
Prediction Time	Datetime (Not Null)	Time Predicted

Route Analysis Table

Field Name	Data Type (Size, Constraints)	Description
Route Id	Integer (Primary Key, Not Null)	Route Identifier
User Id	Integer (Not Null)	User Reference
Source Location	Varchar (150, Not Null)	Starting Point
Destination	Varchar (150, Not Null)	Ending Point
Safe Route Path	Text (Not Null)	Route Details
Risk Level	Varchar (20)	Safe / Moderate / High
Request Time	Datetime (Not Null)	Timestamp

Alerts & Notifications Table

Field Name	Data Type (Size, Constraints)	Description
Alert Id	Integer (Primary Key, Not Null)	Alert Identifier
User Id	Integer (Not Null)	User Reference
Alert Message	Varchar (255, Not Null)	Alert Text
Alert Type	Varchar (50)	Deviation / Risk
Alert Time	Datetime (Not Null)	Alert Timestamp

CHAPTER 7

SYSTEM IMPLEMENTATION

7.1. SOURCE CODE

Packages

```
import math
from flask import Flask, render_template, Response, redirect, request, session, abort, url_for
import mysql.connector
import datetime
from random import randint
from urllib.request import urlopen
import webbrowser
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
```

Login

```
def login():
msg=""
if request.method=='POST':
uname=request.form['uname']
pwd=request.form['pass']
cursor = mydb.cursor()
cursor.execute('SELECT * FROM register WHERE uname = %s AND pass = %s', (uname,
pwd))
account = cursor.fetchone()
if account:
session['username'] = uname
return redirect(url_for('userhome'))
else: msg = 'Incorrect username/password!'
```

User Registration

```
def register():
```

```
msg=""
if request.method=='POST':
name=request.form['name']
mobile=request.form['mobile']
email=request.form['email']
uname=request.form['uname']
pass1=request.form['pass']
mycursor = mydb.cursor()
mycursor.execute("SELECT count(*) FROM register where uname=%s",(uname,))
cnt = mycursor.fetchone()[0]
if cnt==0:
mycursor.execute("SELECT max(id)+1 FROM register")
maxid = mycursor.fetchone()[0]
if maxid is None:
maxid=1
sql = "INSERT INTO register(id,name,mobile,email,uname,pass) VALUES (%s, %s, %s,
%s, %s, %s)"
val = (maxid,name,mobile,email,uname,pass1)
mycursor.execute(sql, val)
mydb.commit()
#print(mycursor.rowcount, "Registered Success")
msg="success"
else:
msg='Already Exist'
```

Training

```
#Preprocessing
def process2():
msg=""
mem=0
cnt=0
cols=0
filename = 'static/dataset/crimes.csv'
data1 = pd.read_csv(filename, header=0)
data2 = list(data1.values.flatten())
```

```
cname=[]
data=[]
dtype=[]
dtt=[]
nv=[]
i=0
sd=len(data1)
rows=len(data1.values)
col=data1.columns
for ss in data1.values:
cnt=len(ss)
i=0
while i<cnt:
j=0
x=0
for rr in data1.values:
dt=type(rr[i])
if rr[i]!="":
x+=1
j+=1
dtt.append(dt)
nv.append(str(x))
i+=1
arr1=np.array(col)
arr2=np.array(nv)
data3=np.vstack((arr1, arr2))
arr3=np.array(data3)
arr4=np.array(dtt)
data=np.vstack((arr3, arr4))
print(data)
cols=cnt
mem=float(rows)*0.75
#Data Analysis
df=pd.read_csv('static/dataset/crimes.csv')
```

```
df.head()
df.drop('Unnamed: 0',axis=1,inplace=True)
df['STATE/UT'].unique()
df.loc[df['STATE/UT'] == 'A&N Islands', 'STATE/UT'] = 'A & N ISLANDS'
df.loc[df['STATE/UT'] == 'D&N Haveli', 'STATE/UT'] = 'D & N HAVELI'
df.loc[df['STATE/UT'] == 'Delhi UT', 'STATE/UT'] = 'DELHI'
df['STATE/UT'] = pd.Series(str.upper(i) for i in df['STATE/UT'])
df['DISTRICT'] = pd.Series(str.upper(i) for i in df['DISTRICT'])
#stroring the sum of all crimes comitted within a state statewise
state_all_crimes = df.groupby('STATE/UT').sum()
#droping the sum of year column
state_all_crimes.drop('Year',axis=1,inplace=True)
#adding a column containig the total crime against women in that state
col_list= list(state_all_crimes)
state_all_crimes['Total']=state_all_crimes[col_list].sum(axis=1)
all_crimes = state_all_crimes
#sorting the statewise crime from highest to lowest
state_all_crimes.sort_values('Total',ascending=False)
state_all_crimes=state_all_crimes.reset_index()
total_df=state_all_crimes.sum(axis=0).reset_index()
tf=pd.DataFrame(total_df)
tf=tf.drop([0])
tf=tf.drop([8])
sorted_df = state_all_crimes.sort_values('Total',ascending=False)
#Feature Extraction
state_all_crimes=state_all_crimes.reset_index()
total_df=state_all_crimes.sum(axis=0).reset_index()
tf=pd.DataFrame(total_df)
tf=tf.drop([0])
tf=tf.drop([8])
sorted_df = state_all_crimes.sort_values('Total',ascending=False)
dat=all_crimes = all_crimes.reset_index()
for ss in dat.values:
data.append(ss)
```

```
all_crimes.shape
#finding the mean number of crimes
m=all_crimes['Total'].mean()
print('mean=',m)
#finding the quantiles
q = np.quantile(all_crimes['Total'],[0.25,0.75])
print(q)
l=q[0]
u=q[1]
#copying the state_all_crimes to a new dataframe to normalise values and predict
df_kmeans = all_crimes.loc[:,all_crimes.columns!="STATE/UT"]
#adding an additional column called output
output=[]
for i in df_kmeans['Total']:
if i >= m:
output.append(1)#redzone
elif m > i:
output.append(0)#safe
all_crimes['output']=output
df_kmeans_y=all_crimes['output']
#feature scaling
from sklearn.preprocessing import MinMaxScaler
cols = df_kmeans.columns
ms=MinMaxScaler()
df_kmeans = ms.fit_transform(df_kmeans)
df_kmeans = pd.DataFrame(df_kmeans,columns=[cols])
df_kmeans.head()
from sklearn.cluster import KMeans
kmeans = KMeans(n_clusters=2, random_state=0)
kmeans.fit(df_kmeans)
kmeans.inertia_
#checking the accuracy
labels = kmeans.labels_
# check how many of the samples were correctly labeled
```

```
correct_labels = sum(df_kmeans_y == labels)
print('labels:',labels)
print('df_kmeans output:',df_kmeans_y)
print("Result: %d out of %d samples were correctly labeled." % (correct_labels,
df_kmeans_y.size))
#based on the prediction of the k means algorithm classifying the states
#as safe or unsafe for women
final=[]
for i in range(len(labels)):
state=all_crimes['STATE/UT'][i]
label = labels[i]
if label == 1:
final.append([state,'unsafe'])
else:
final.append([state,'safe'])
dat2=final_df = pd.DataFrame(final, columns=['STATES/UT', 'SAFE/UNSAFE'])
#final_df
for ss2 in dat2.values:
data2.append(ss2)
#SVM Classification
class SVM:
def fit(self, X, y):
n_samples, n_features = X.shape# P = X^T X
K = np.zeros((n_samples, n_samples))
for i in range(n_samples):
for j in range(n_samples):
K[i,j] = np.dot(X[i], X[j])
P = cvxopt.matrix(np.outer(y, y) * K)# q = -1 (1xN)
q = cvxopt.matrix(np.ones(n_samples) * -1)# A = y^T
A = cvxopt.matrix(y, (1, n_samples))# b = 0
b = cvxopt.matrix(0.0)# -1 (NxN)
G = cvxopt.matrix(np.diag(np.ones(n_samples) * -1))# 0 (1xN)
h = cvxopt.matrix(np.zeros(n_samples))
solution = cvxopt.solvers.qp(P, q, G, h, A, b)# Lagrange multipliers
```

```
a = np.ravel(solution['x'])# Lagrange have non zero lagrange multipliers
sv = a > 1e-5
ind = np.arange(len(a))[sv]
self.a = a[sv]
self.sv = X[sv]
self.sv_y = y[sv]# Intercept
self.b = 0
for n in range(len(self.a)):
self.b += self.sv_y[n]
self.b -= np.sum(self.a * self.sv_y * K[ind[n], sv])
self.b /= len(self.a)# Weights
self.w = np.zeros(n_features)
for n in range(len(self.a)):
self.w += self.a[n] * self.sv_y[n] * self.sv[n]
def project(self, X):
return np.dot(X, self.w) + self.b
def predict(self, X):
return np.sign(self.project(X))
df['STATE/UT'].unique()
df.loc[df['STATE/UT'] == 'A&N Islands', 'STATE/UT'] = 'A & N ISLANDS'
df.loc[df['STATE/UT'] == 'D&N Haveli', 'STATE/UT'] = 'D & N HAVELI'
df.loc[df['STATE/UT'] == 'Delhi UT', 'STATE/UT'] = 'DELHI'
#converting all the state names to capitals
df['STATE/UT'] = pd.Series(str.upper(i) for i in df['STATE/UT'])
df['DISTRICT'] = pd.Series(str.upper(i) for i in df['DISTRICT'])
#stroring the sum of all crimes comitted within a state statewise
state_all_crimes = df.groupby('STATE/UT').sum()
#droping the sum of year column
state_all_crimes.drop('Year',axis=1,inplace=True)
#adding a column containig the total crime against women in that state
col_list= list(state_all_crimes)
state_all_crimes['Total']=state_all_crimes[col_list].sum(axis=1)
all_crimes = state_all_crimes
#sorting the statewise crime from highest to lowest
```

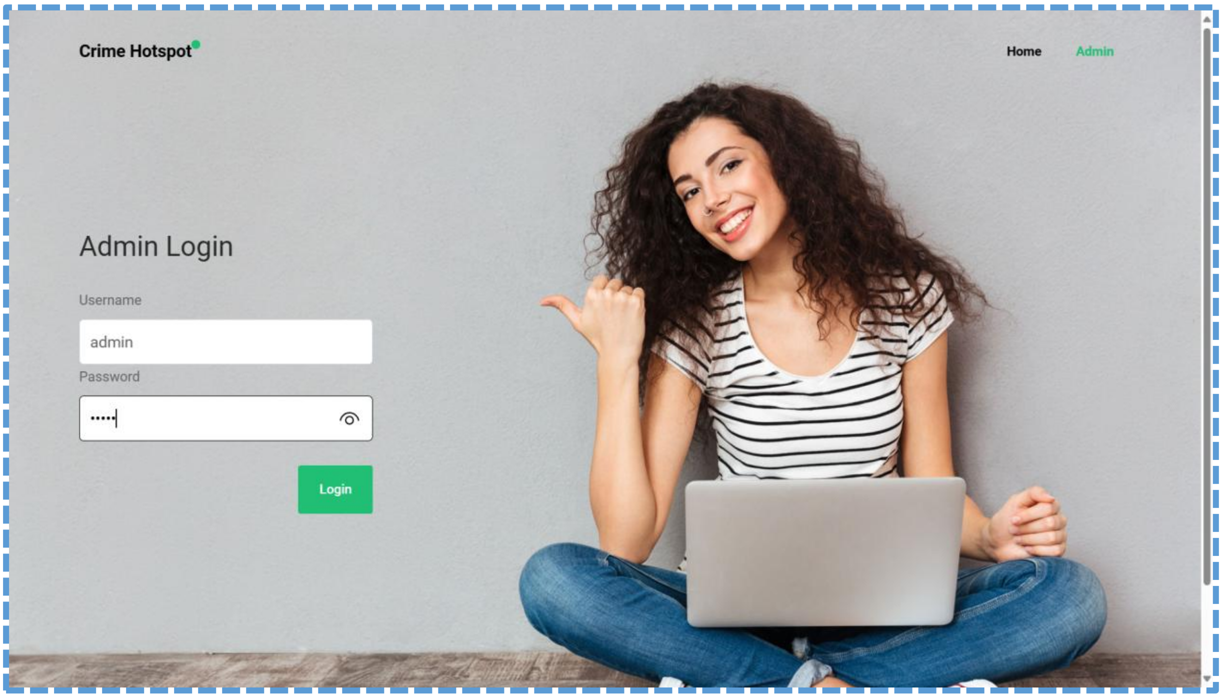
```
state_all_crimes.sort_values('Total',ascending=False)
state_all_crimes=state_all_crimes.reset_index()
total_df=state_all_crimes.sum(axis=0).reset_index()
tf=pd.DataFrame(total_df)
tf=tf.drop([0])
sorted_df = state_all_crimes.sort_values('Total',ascending=False)
dat=all_crimes = all_crimes.reset_index()
for ss in dat.values:
dt=[]
dt.append(ss[0])
f1=float(ss[8])
if f1>200000:
dt.append("Not Safety")
elif f1>100000:
dt.append("Medium Safety")
elif f1>50000:
dt.append("Low Safety")
else:
dt.append("Safety")
data.append(dt)
```

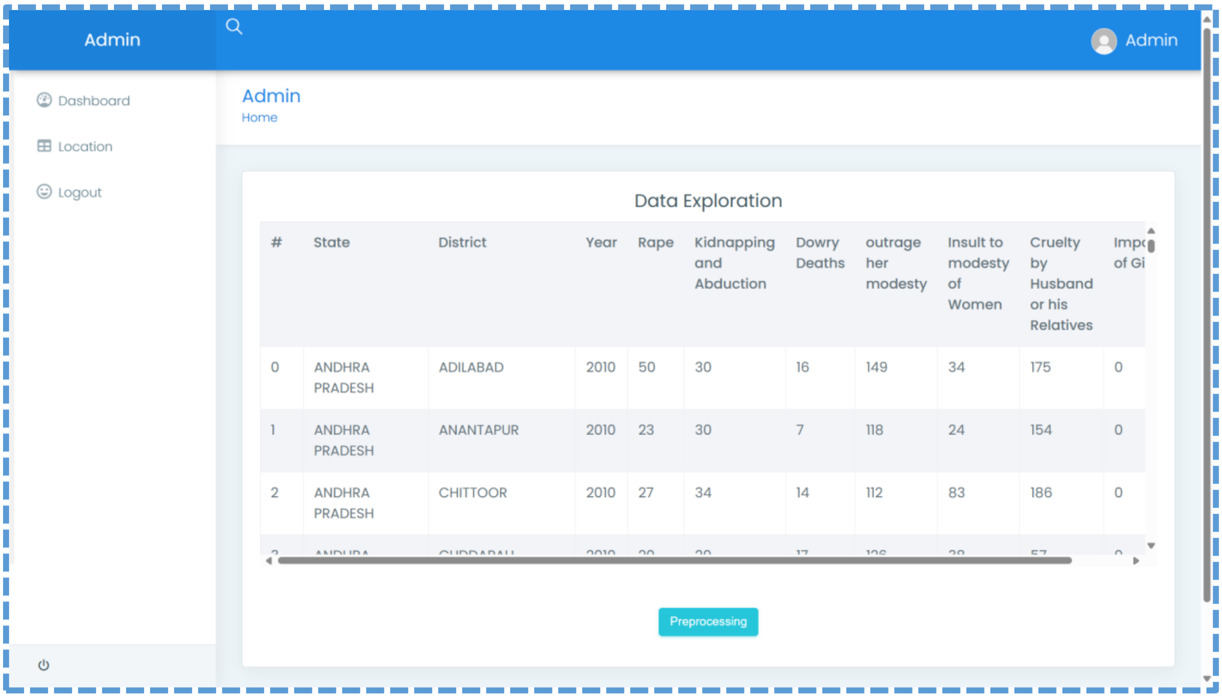
Search Location

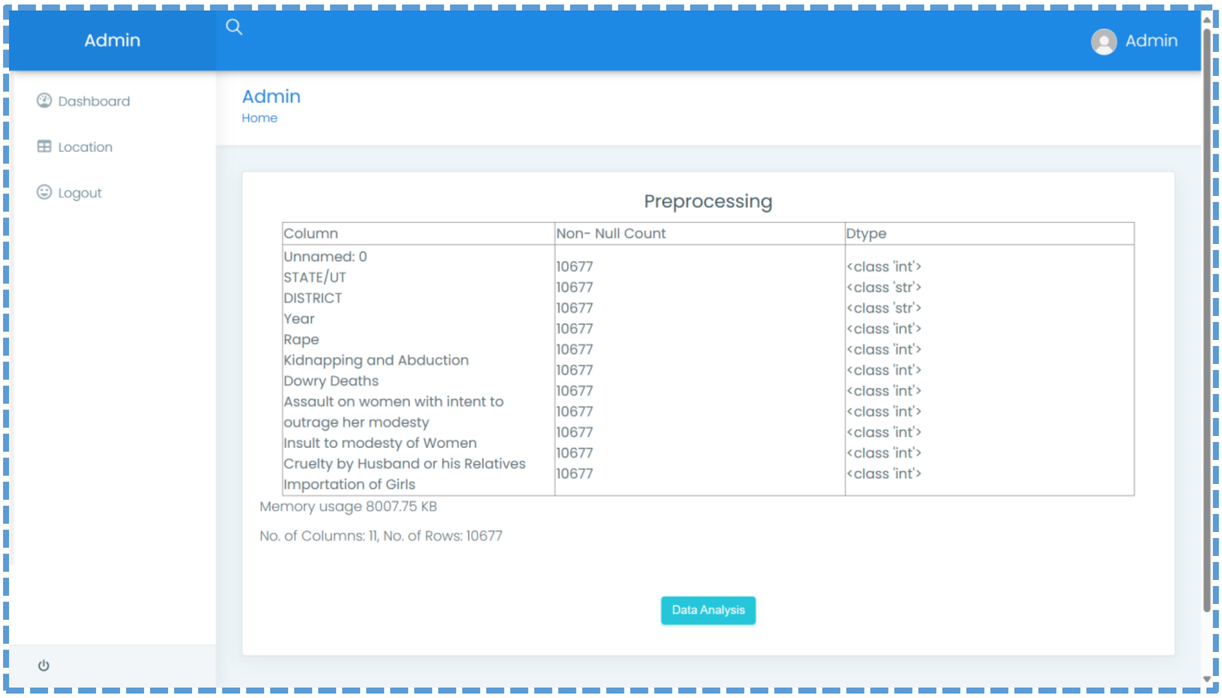
```
def userhome():
msg=""
uname=""
st=""
if 'username' in session:
uname = session['username']
mycursor = mydb.cursor()
mycursor.execute("SELECT * FROM register where uname=%s",(uname,))
usr = mycursor.fetchone()
if request.method=='POST':
splace=request.form['splace']
dplace=request.form['dplace']
loc1='%'+splace+'%'
```

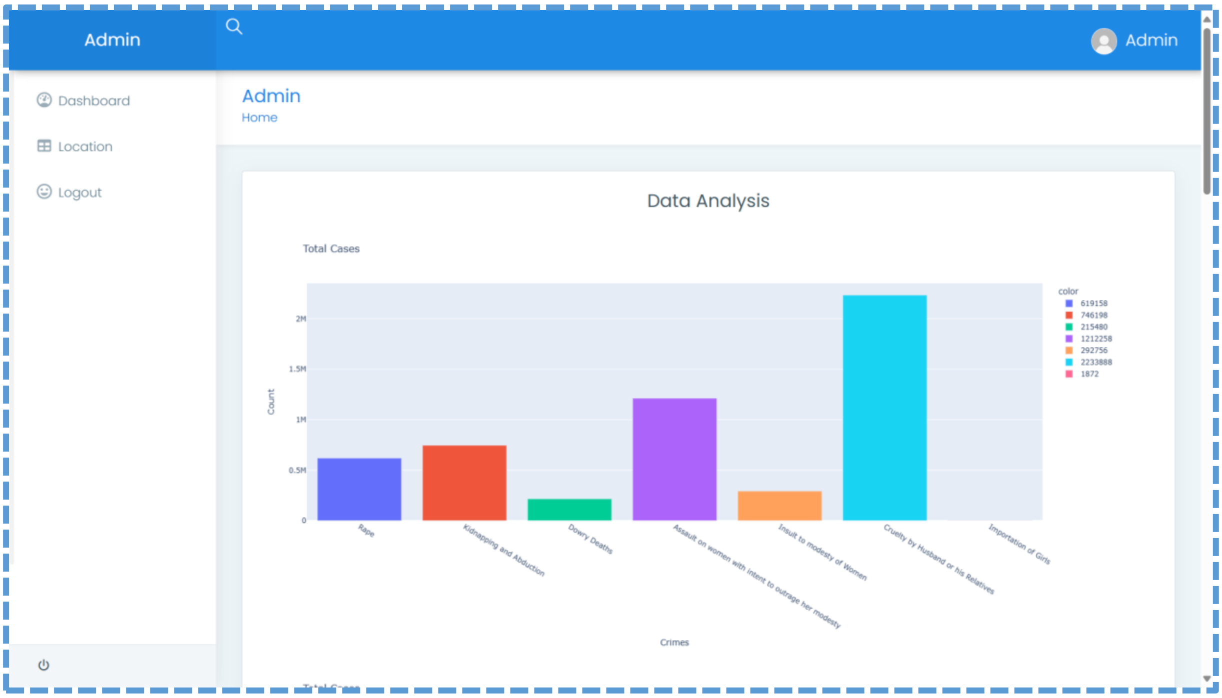
```
loc2='%'+dplace+'%'
mycursor.execute("SELECT count(*) FROM routes where location like %s",(loc1,))
cn = mycursor.fetchone()[0]
if cn>0:
mycursor.execute("SELECT * FROM routes where location like %s",(loc1,))
dd1 = mycursor.fetchall()
v2=""
st="1"
for dd in dd1:
rid=dd[2]
sid=dd[3]
mycursor.execute("SELECT count(*) FROM routes where location like %s &&
route_id=%s",(loc2,rid))
cn2 = mycursor.fetchone()[0]
if cn2>0:
mycursor.execute("SELECT * FROM routes where location like %s &&
route_id=%s",(loc2,rid))
dd2 = mycursor.fetchall()
for dd3 in dd2:
rid2=dd3[2]
sid2=dd3[3]
if sid<sid2:
v=str(sid)+","+str(sid2)
else:
v=str(sid2)+","+str(sid)
v2=str(rid2)+","+v+"|"
ff=open("static/route.txt","w")
ff.write(v2)
ff.close()
msg="Location"
else: st="2
```

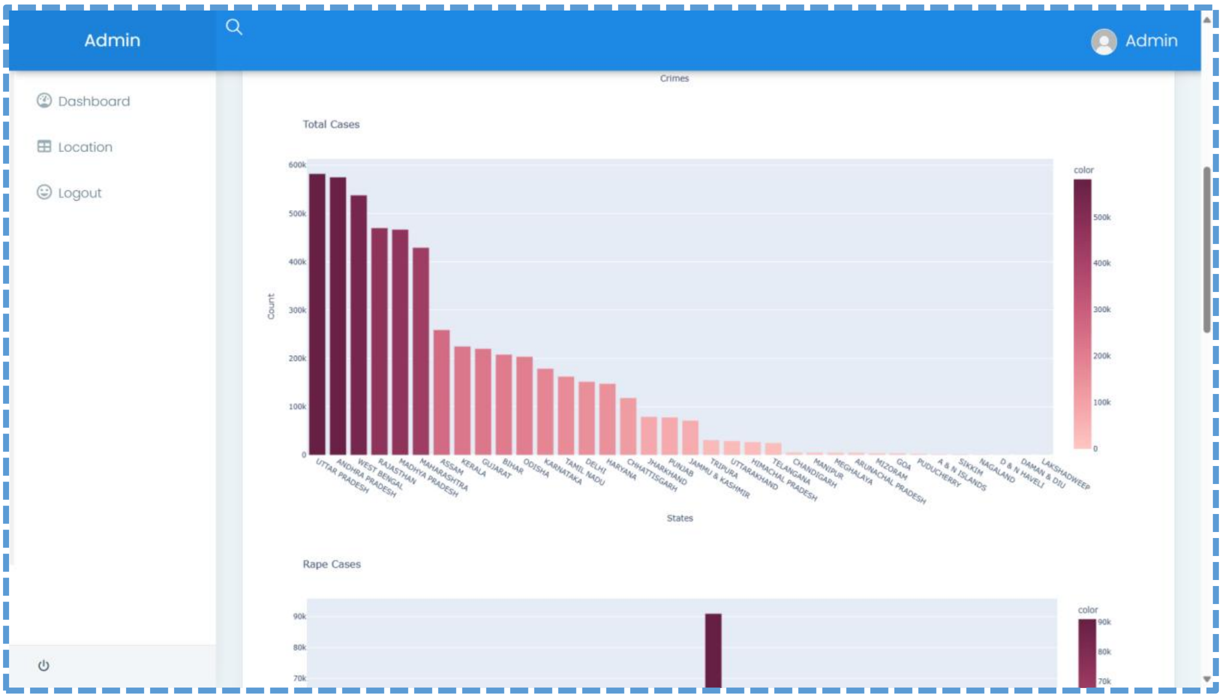
7.2. SCREENSHOTS



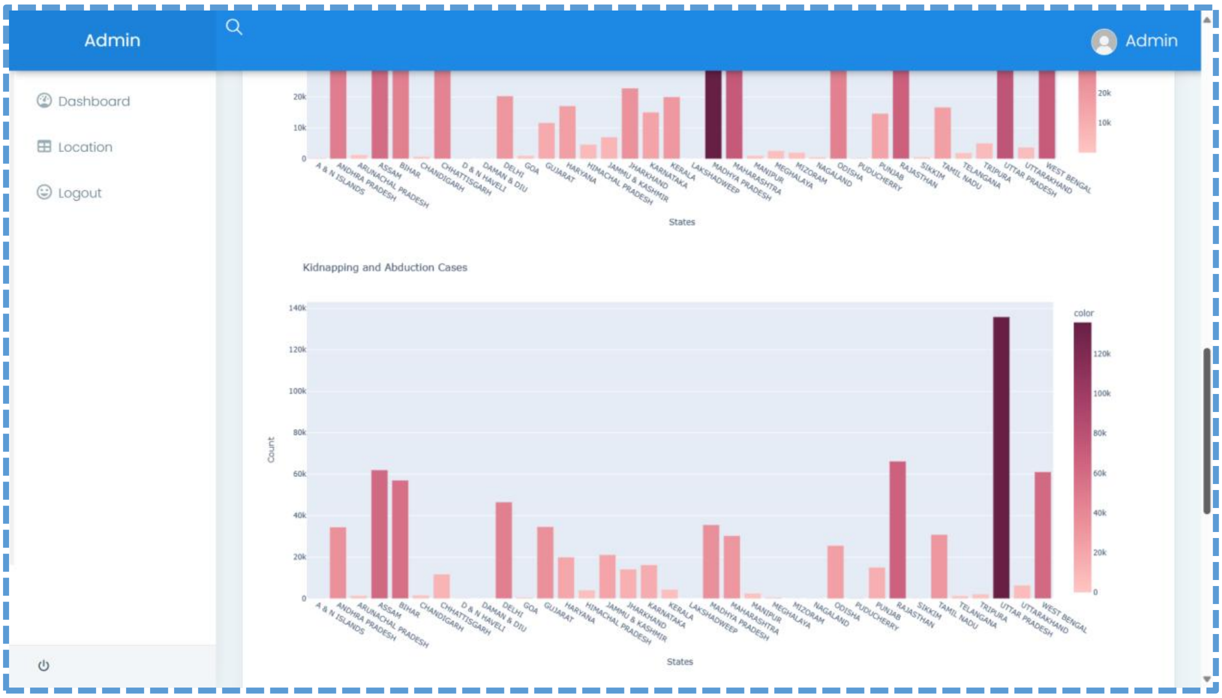



Admin
Admin
Dashboard
Location
Logout
Admin
Home
Preprocessing
Column
Non- Null Count
Dtype
Unnamed: 0
STATE/UT
DISTRICT
Year
Rape
Kidnapping and Abduction
Dowry Deaths
Assault on women with intent to outrage her modesty
Insult to modesty of Women
Cruelty by Husband or his Relatives
Importation of Girls
10677
<class 'int'>
<class 'str'>
Memory usage 8007.75 KB
No. of Columns: 11, No. of Rows: 10677
Data Analysis

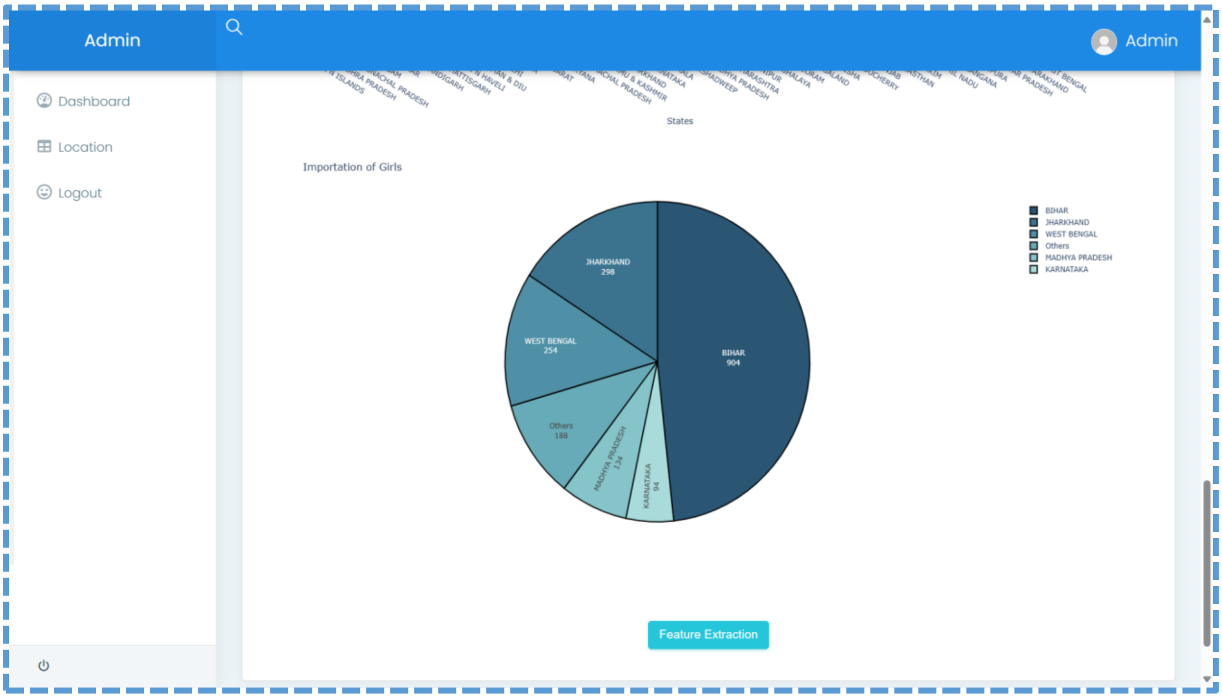

Admin
Admin
Dashboard
Location
Logout
Admin
Home
Data Analysis
Total Cases
color
619158
746198
215480
1212258
292756
2233888
1872
2M
1.5M
1M
0.5M
0
Count
Rape
Kidnapping and Abduction
Dowry Deaths
Assault on women with intent to outrage her modesty
Insult to modesty of Women
Cruelty by Husband or his Relatives
Importation of Girls
Crimes


Admin
Admin
Dashboard
Location
Logout
Crimes
Total Cases
Count
States
color
Rape Cases

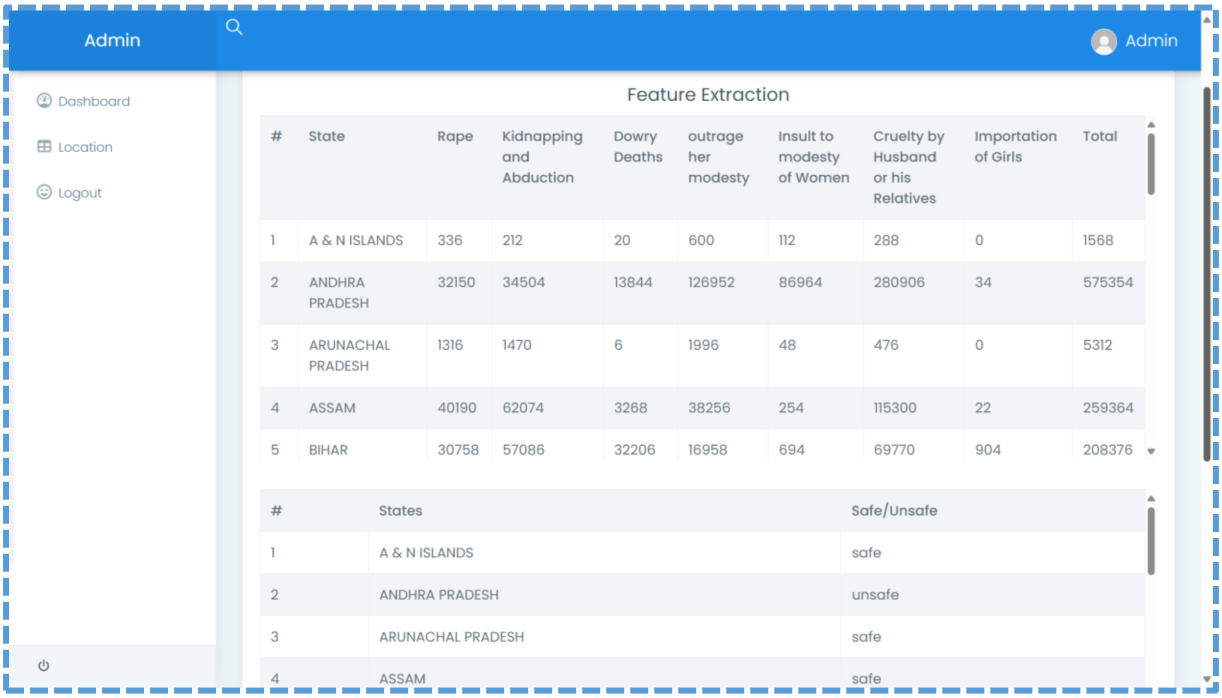

Admin
Admin
Dashboard
Location
Logout
Count
States
color
Kidnapping and Abduction Cases



Admin
Admin
Dashboard
Location
Logout
States
Kidnapping and Abduction Cases
Count
color
States

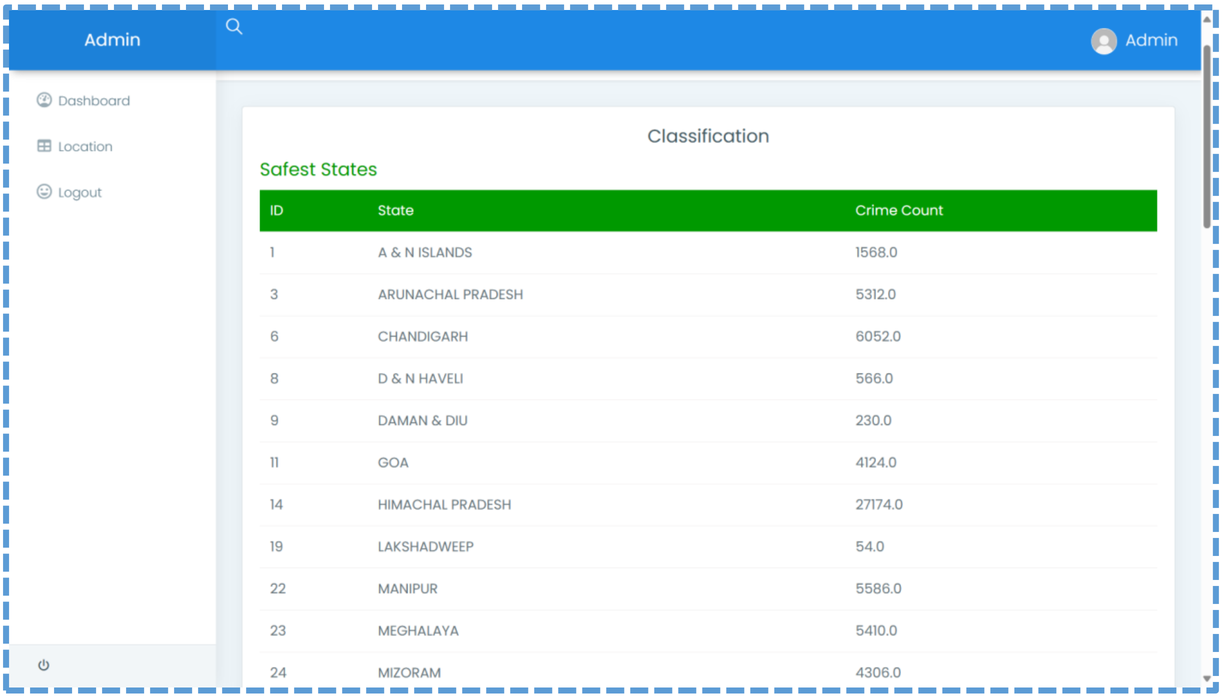


Admin
Admin
Dashboard
Location
Logout
States
Importation of Girls
JHARKHAND
298
WEST BENGAL
254
Others
188
MADHYA PRADESH
134
KARNATAKA
94
BIHAR
904
BIHAR
JHARKHAND
WEST BENGAL
Others
MADHYA PRADESH
KARNATAKA
Feature Extraction



#	State	Rape	Kidnapping and Abduction	Dowry Deaths	outrage her modesty	Insult to modesty of Women	Cruelty by Husband or his Relatives	Importation of Girls	Total
1	A & N ISLANDS	336	212	20	600	112	288	0	1568
2	ANDHRA PRADESH	32150	34504	13844	126952	86964	280906	34	575354
3	ARUNACHAL PRADESH	1316	1470	6	1996	48	476	0	5312
4	ASSAM	40190	62074	3268	38256	254	115300	22	259364
5	BIHAR	30758	57086	32206	16958	694	69770	904	208376

#	States	Safe/Unsafe
1	A & N ISLANDS	safe
2	ANDHRA PRADESH	unsafe
3	ARUNACHAL PRADESH	safe
4	ASSAM	safe



ID	State	Crime Count
1	A & N ISLANDS	1568.0
3	ARUNACHAL PRADESH	5312.0
6	CHANDIGARH	6052.0
8	D & N HAVELI	566.0
9	DAMAN & DIU	230.0
11	GOA	4124.0
14	HIMACHAL PRADESH	27174.0
19	LAKSHADWEEP	54.0
22	MANIPUR	5586.0
23	MEGHALAYA	5410.0
24	MIZORAM	4306.0

States in Risk (Unsafe)

ID	State	Crime Count
15	JAMMU & KASHMIR	71692.0
16	JHARKHAND	79580.0
28	PUNJAB	78258.0

States in Dangerous (Unsafe)

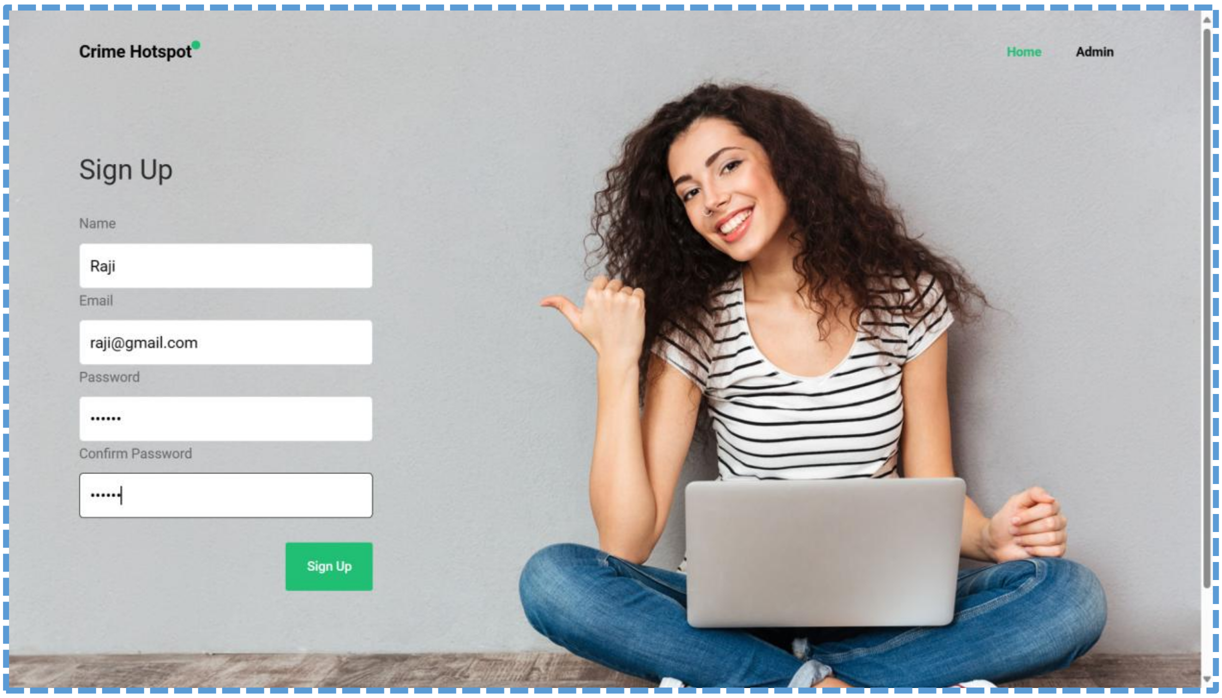
ID	State	Crime Count
7	CHHATTISGARH	118386.0
10	DELHI	152162.0
13	HARYANA	147924.0
17	KARNATAKA	179160.0
31	TAMIL NADU	162756.0

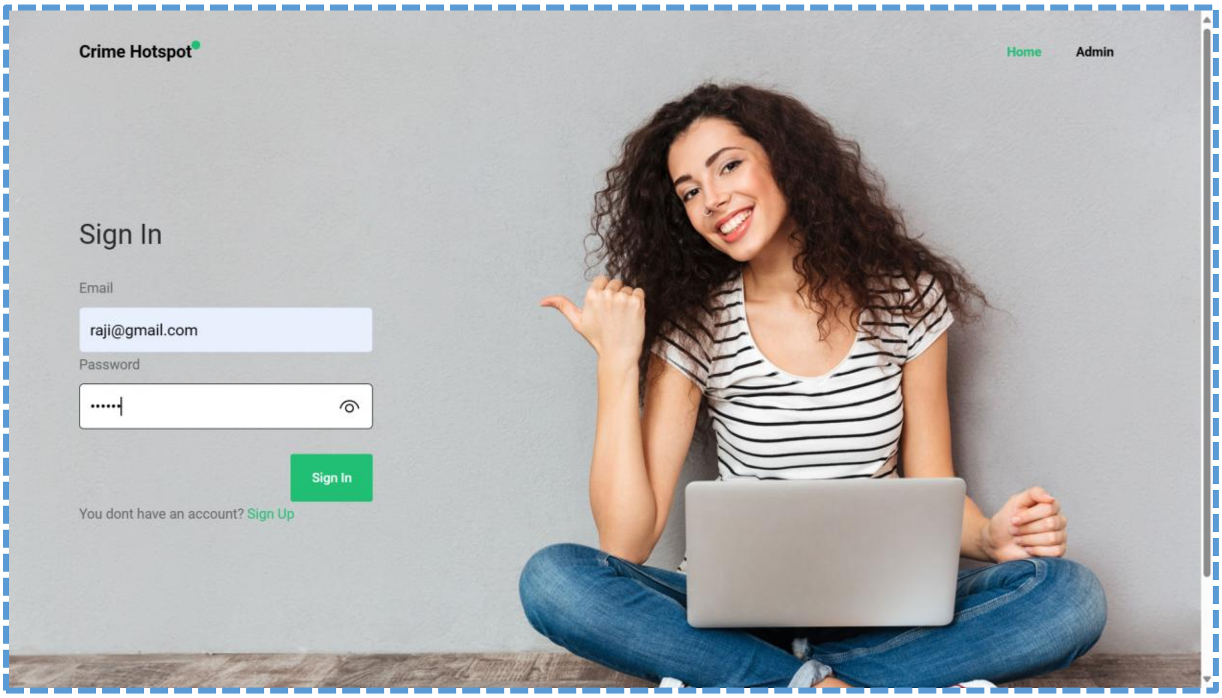
States in Extreme Risk (Unsafe)

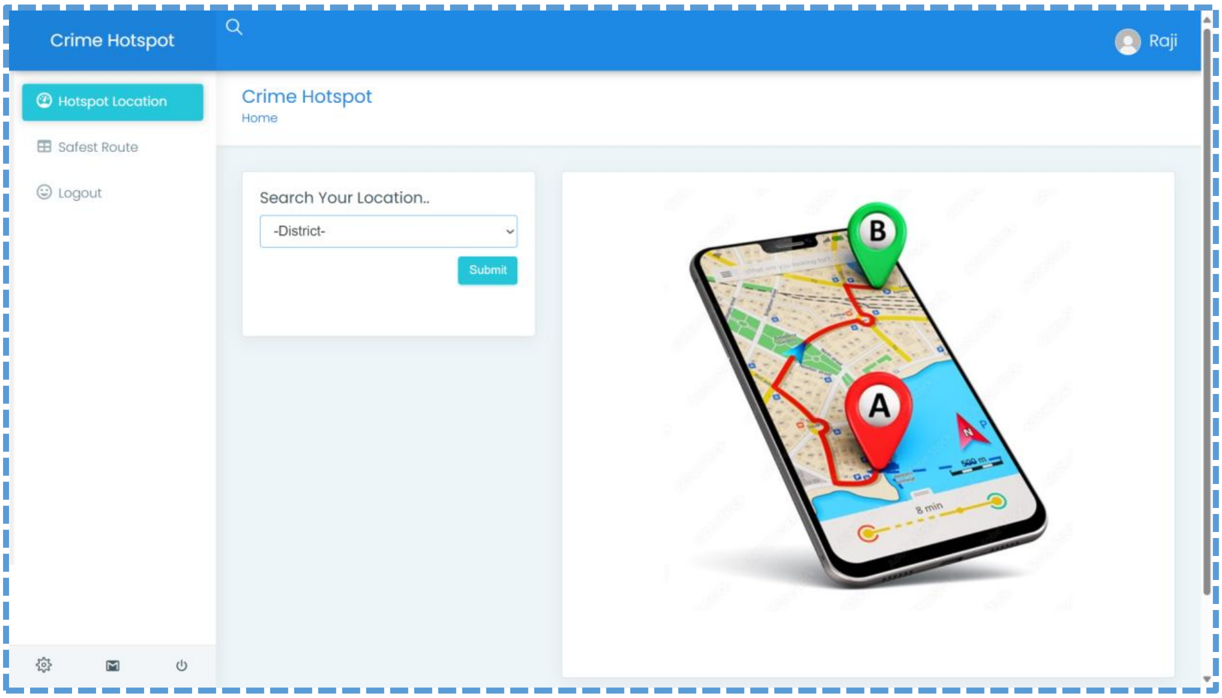
31	TAMIL NADU	162756.0

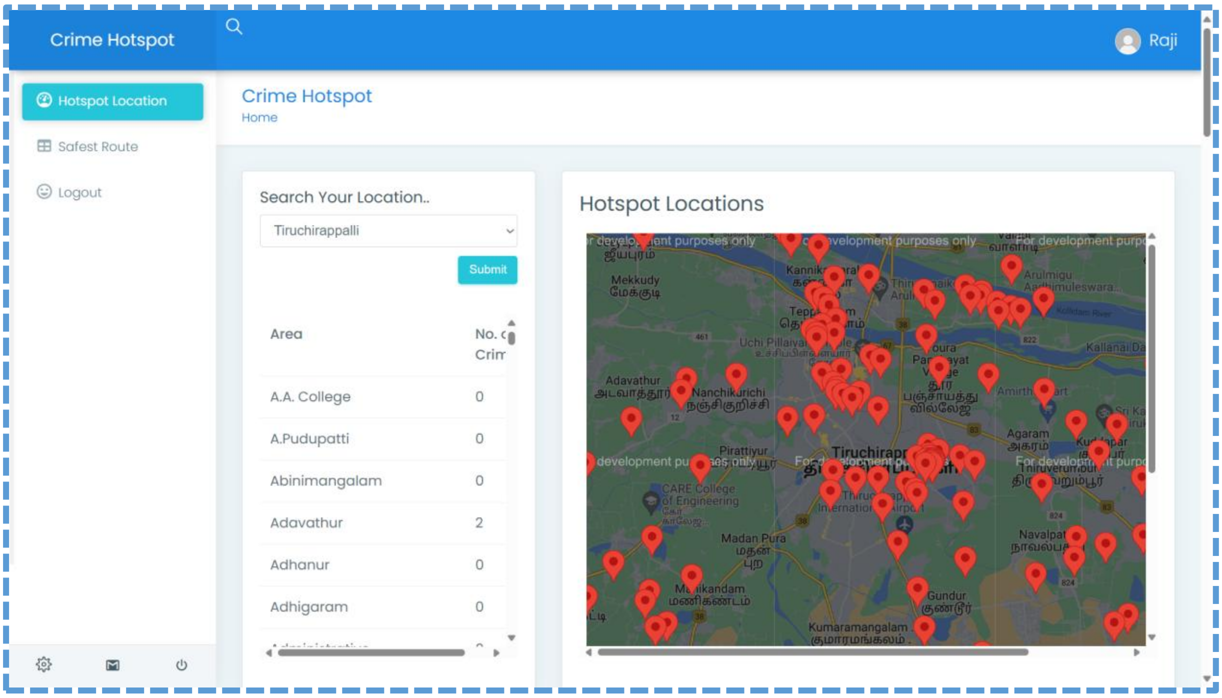
States in Extreme Risk (Unsafe)

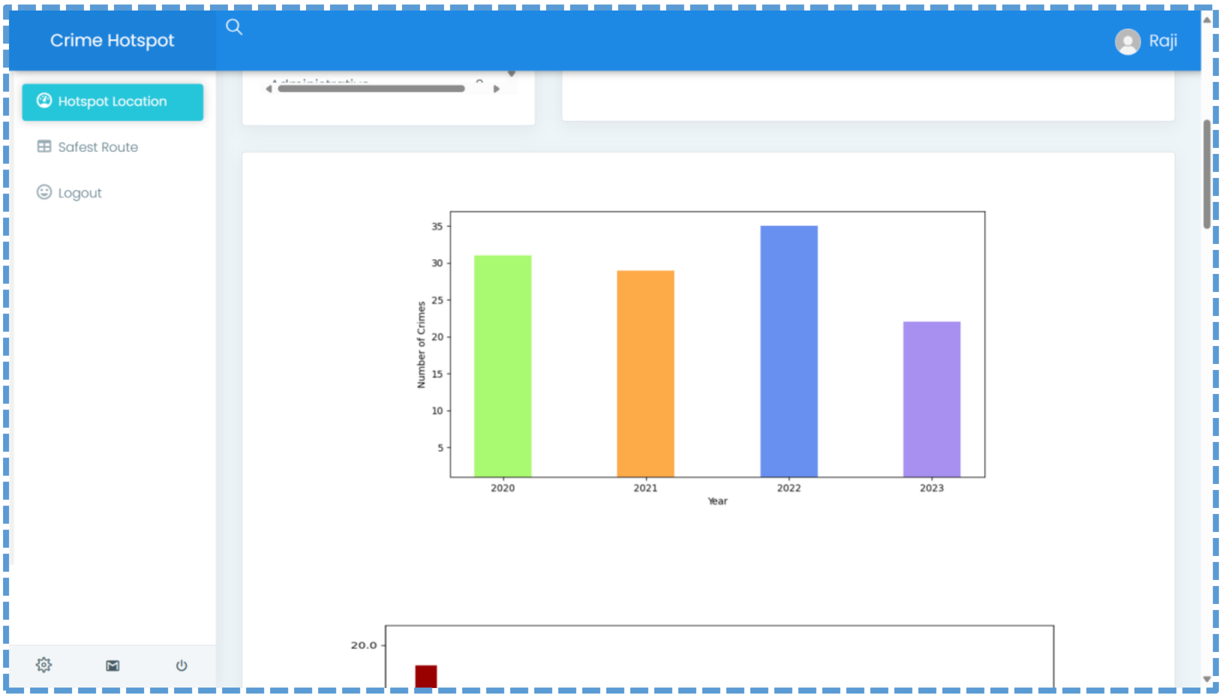
ID	State	Crime Count
2	ANDHRA PRADESH	575354.0
4	ASSAM	259364.0
5	BIHAR	208376.0
12	GUJARAT	220306.0
18	KERALA	225176.0
20	MADHYA PRADESH	467086.0
21	MAHARASHTRA	429584.0
26	ODISHA	203794.0
29	RAJASTHAN	470186.0
34	UTTAR PRADESH	582398.0
36	WEST BENGAL	537976.0

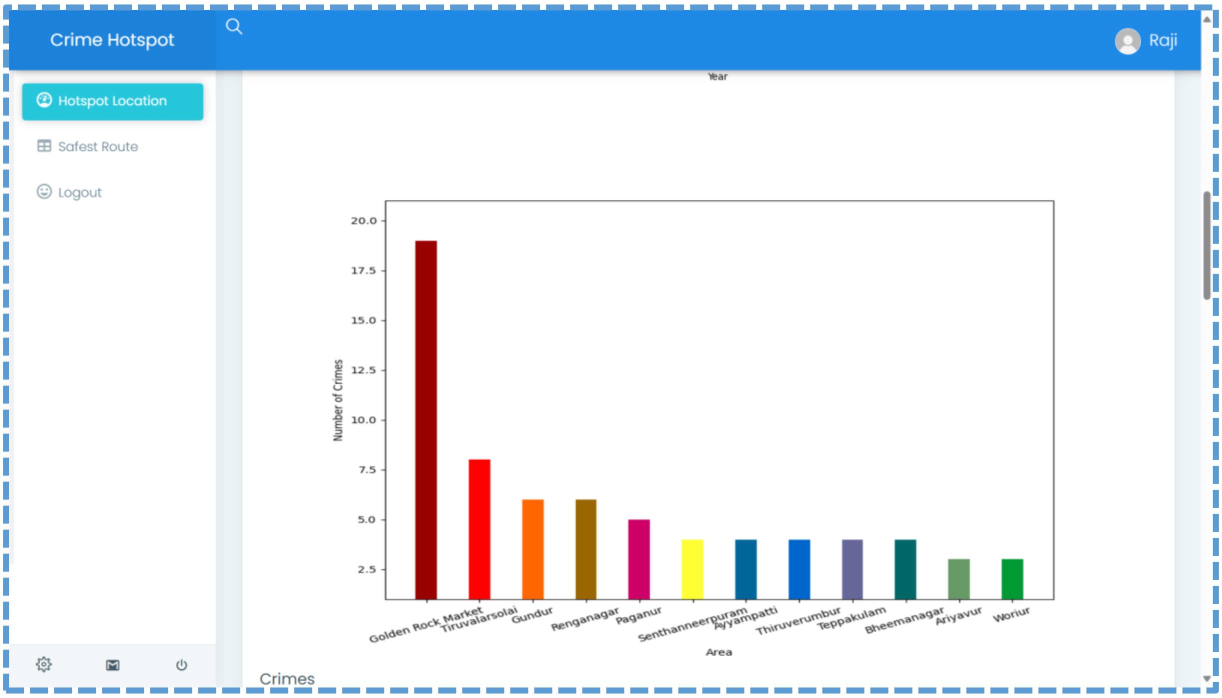

Crime Hotspot
Home
Admin
Sign Up
Name
Raji
Email
raji@gmail.com
Password
Confirm Password
Sign Up

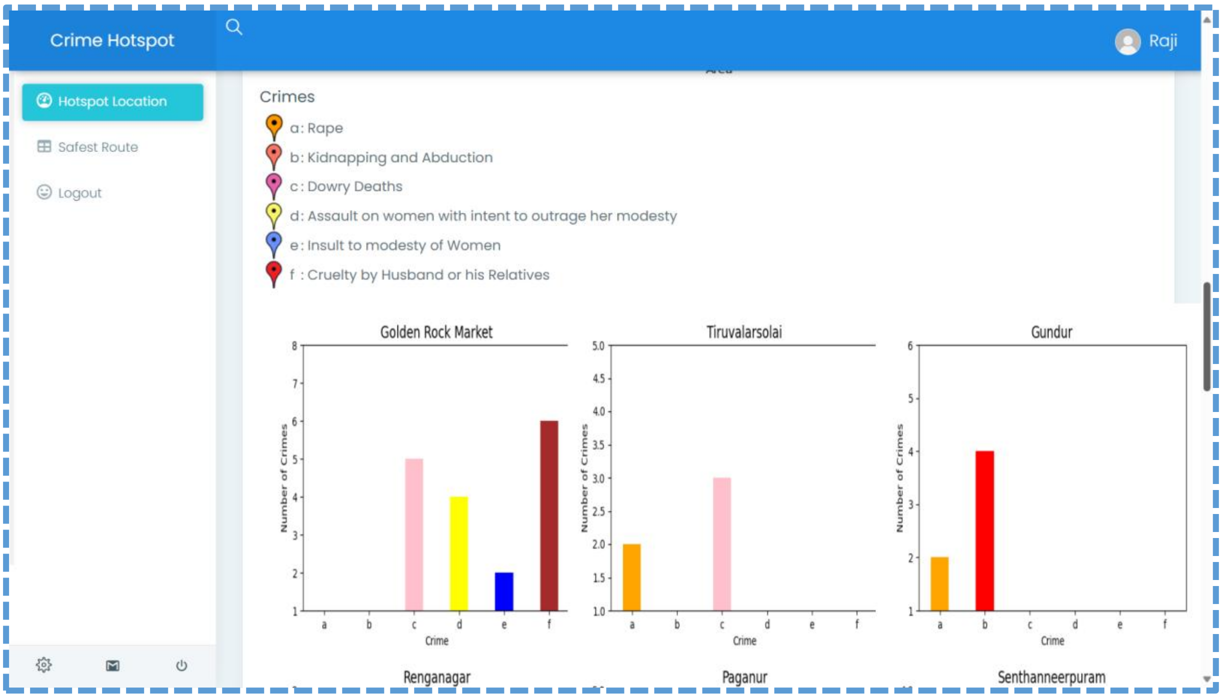

Crime Hotspot
Home
Admin
Sign In
Email
raji@gmail.com
Password
Sign In
You dont have an account? Sign Up

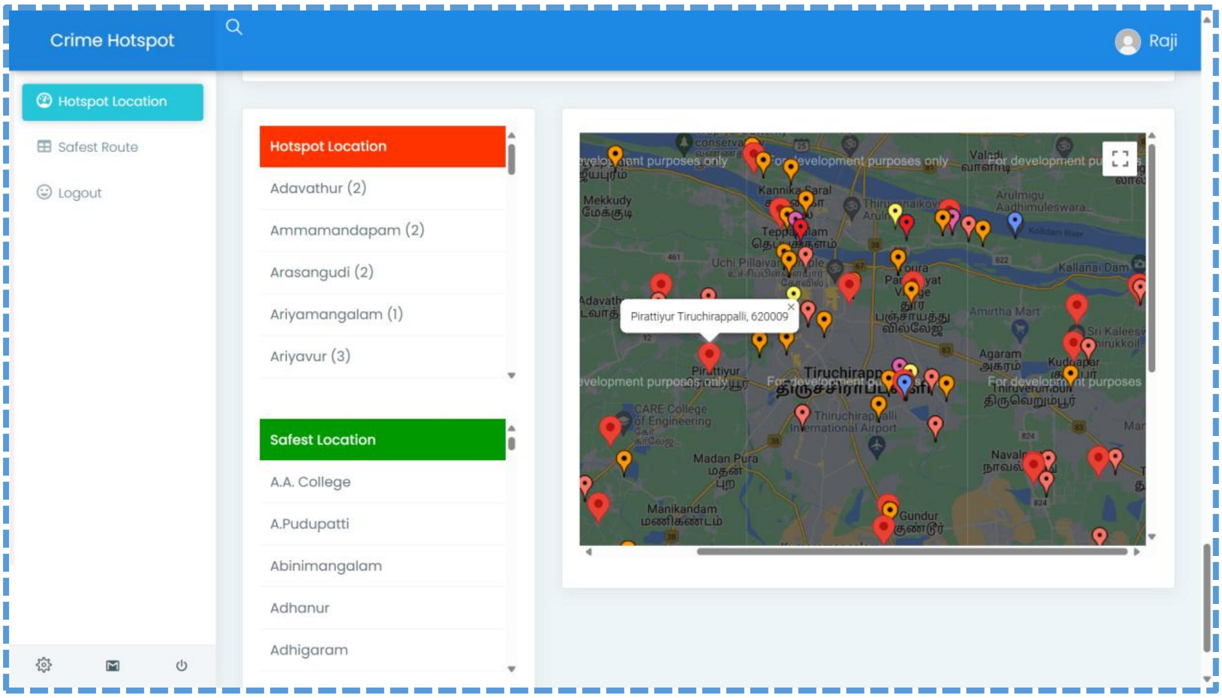

Crime Hotspot
Raji
Hotspot Location
Safest Route
Logout
Crime Hotspot
Home
Search Your Location..
-District-
Submit

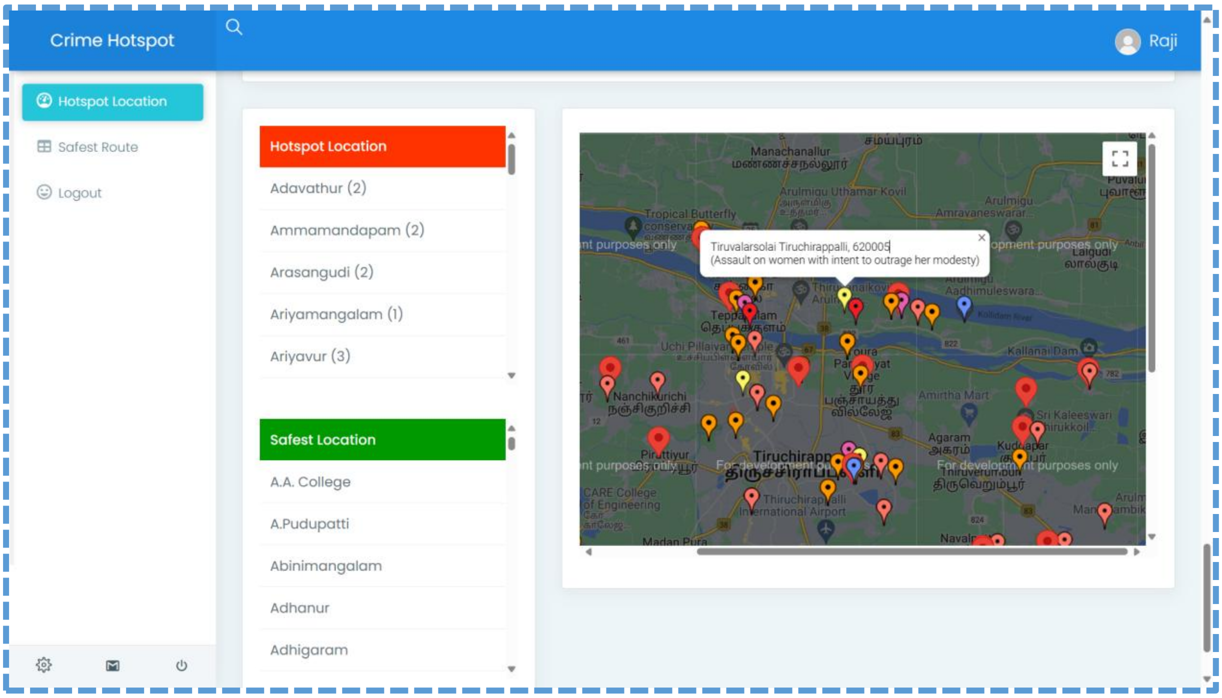

Crime Hotspot
Raji
Hotspot Location
Safest Route
Logout
Crime Hotspot
Home
Search Your Location..
Tiruchirappalli
Submit
Area
A.A. College 0
A.Pudupatti 0
Abinimangalam 0
Adavathur 2
Adhanur 0
Adhigaram 0
Hotspot Locations

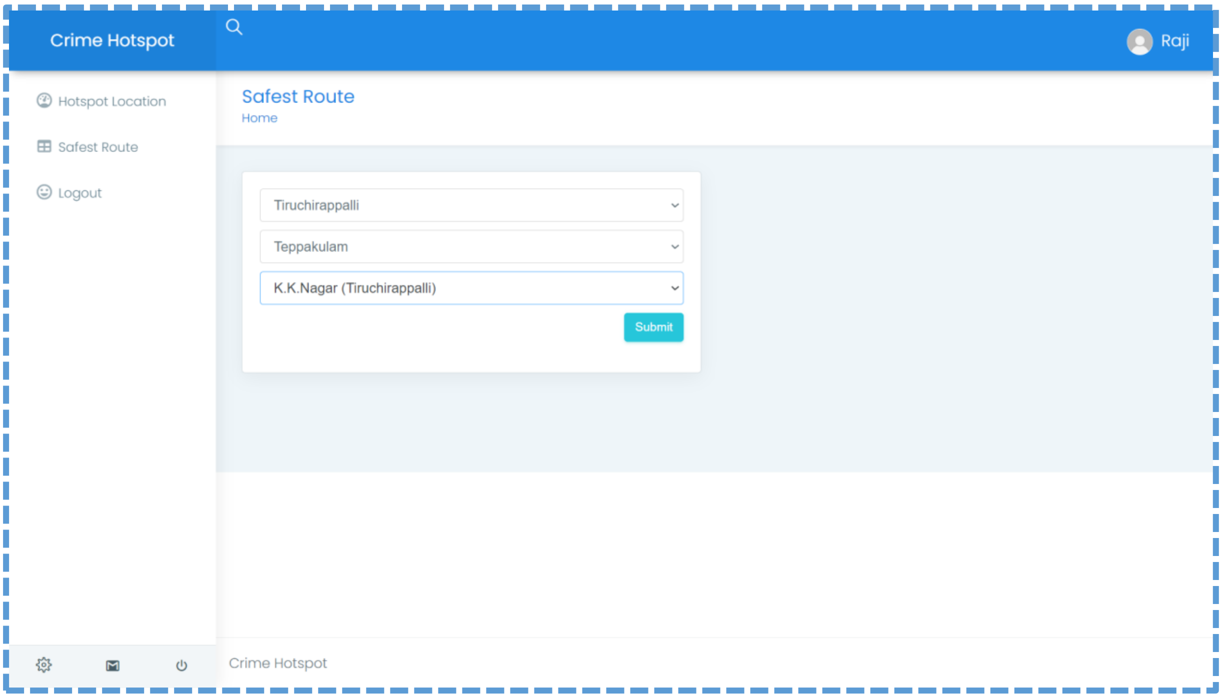

Crime Hotspot
Raji
Hotspot Location
Safest Route
Logout
Number of Crimes
Year
2020
2021
2022
2023

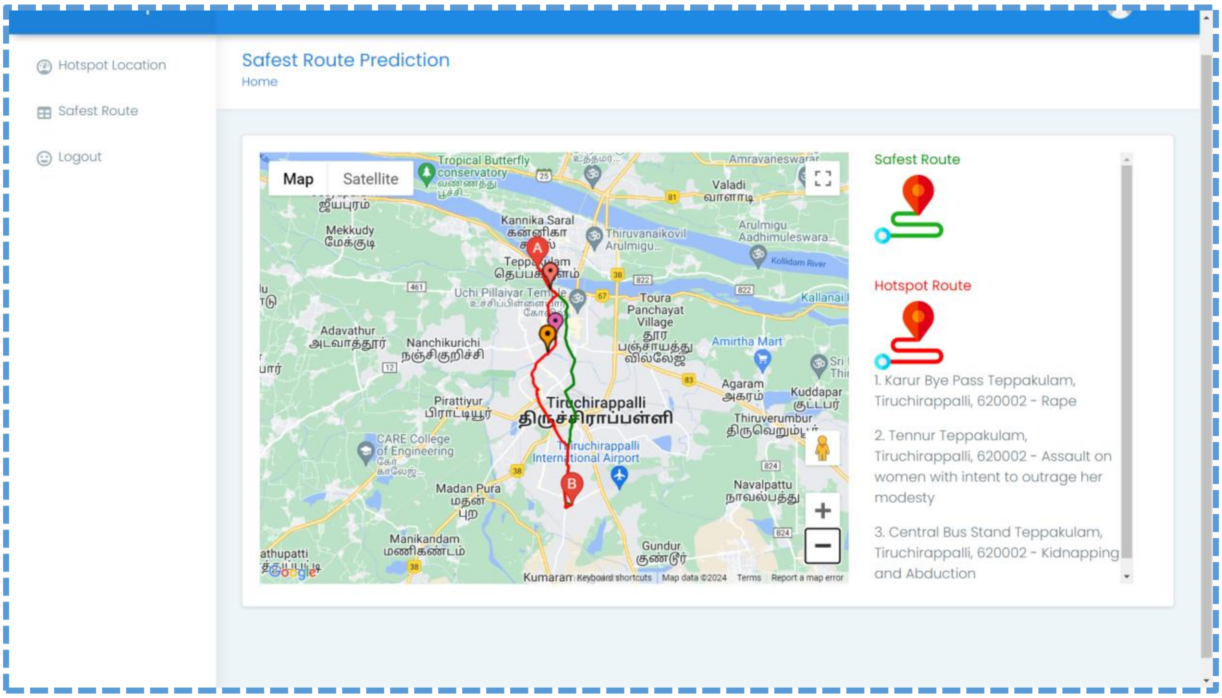

Crime Hotspot
Raji
Hotspot Location
Safest Route
Logout
Number of Crimes
Area
Golden Rock Market
Tiruvalarsolai
Gundur
Renganagar
Paganur
Senthanneerpuram
Ayyampatti
Thiruverumbur
Teppakulam
Bheemanagar
Ariyavur
Woriur
Crimes

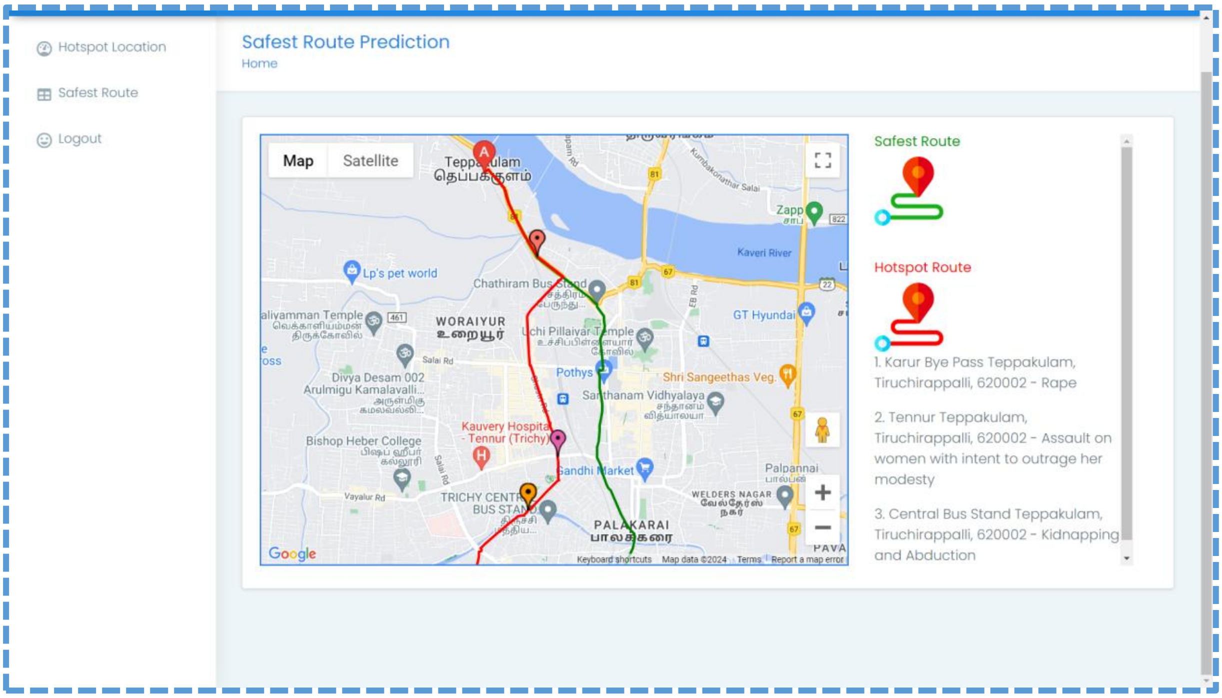

Crime Hotspot
Raji
Hotspot Location
Safest Route
Logout
Crimes
a: Rape
b: Kidnapping and Abduction
c: Dowry Deaths
d: Assault on women with intent to outrage her modesty
e: Insult to modesty of Women
f : Cruelty by Husband or his Relatives
Golden Rock Market
Tiruvalarsolai
Gundur
Number of Crimes
Crime
Renganagar
Paganur
Senthanneerpuram


Crime Hotspot
Raji
Hotspot Location
Safest Route
Logout
Hotspot Location
Adavathur (2)
Ammamandapam (2)
Arasangudi (2)
Ariyamangalam (1)
Ariyavur (3)
Safest Location
A.A. College
A.Pudupatti
Abinimangalam
Adhanur
Adhigaram
Pirattiyur Tiruchirappalli, 620009


Crime Hotspot
Raji
Hotspot Location
Safest Route
Logout
Hotspot Location
Adavathur (2)
Ammamandapam (2)
Arasangudi (2)
Ariyamangalam (1)
Ariyavur (3)
Safest Location
A.A. College
A.Pudupatti
Abinimangalam
Adhanur
Adhigaram
Tiruvalarsolai Tiruchirappalli, 620005
(Assault on women with intent to outrage her modesty)


Crime Hotspot
Raji
Hotspot Location
Safest Route
Logout
Safest Route
Home
Tiruchirappalli
Teppakulam
K.K.Nagar (Tiruchirappalli)
Submit
Crime Hotspot


Hotspot Location
Safest Route
Logout
Safest Route Prediction
Home
Map
Satellite
Safest Route
Hotspot Route
1. Karur Bye Pass Teppakulam, Tiruchirappalli, 620002 - Rape
2. Tennur Teppakulam, Tiruchirappalli, 620002 - Assault on women with intent to outrage her modesty
3. Central Bus Stand Teppakulam, Tiruchirappalli, 620002 - Kidnapping and Abduction


Hotspot Location
Safest Route
Logout
Safest Route Prediction
Home
Map
Satellite
Safest Route
Hotspot Route
1. Karur Bye Pass Teppakulam, Tiruchirappalli, 620002 - Rape
2. Tennur Teppakulam, Tiruchirappalli, 620002 - Assault on women with intent to outrage her modesty
3. Central Bus Stand Teppakulam, Tiruchirappalli, 620002 - Kidnapping and Abduction

CHAPTER 8
SYSTEM TESTING

8.1. SOFTWARE TESTING

Testing is a crucial part of the software development life cycle, ensuring that the system functions correctly, meets specifications, and provides a reliable and safe experience to users. For the web-based project for predictive crime zone mapping, route optimization, and safety alerts for women, the following types of testing are relevant:

- **Unit Testing:** Each individual module, such as the ST-GCN Crime Hotspot Predictor, Route Analysis module, and Notification system, is tested independently to verify that it performs its intended function correctly. This helps identify and fix errors at an early stage.
- **Integration Testing:** This ensures that all modules work together seamlessly. For instance, the integration between crime prediction, route evaluation, and real-time alert modules is tested to confirm proper communication, data flow, and synchronization.
- **System Testing:** The complete system is tested as a whole to validate that it meets both functional and non-functional requirements, including hotspot prediction accuracy, route safety evaluation, timely notifications, and visualization on maps.
- **Acceptance Testing:** End-users test the system to confirm it satisfies their needs, such as receiving accurate route suggestions, real-time alerts about high-risk areas, and clear visualization of crime hotspots.
- **Performance Testing:** The system is evaluated under different load conditions to assess speed, scalability, stability, and responsiveness. This includes handling multiple user requests simultaneously, processing real-time crime data, and rendering maps efficiently.
- **User Acceptance Testing (UAT):** Conducted in a simulated environment with real users to ensure the system is intuitive, responsive, and ready for deployment. UAT validates that the system effectively enhances personal safety, supports informed travel decisions, and assists law enforcement in proactive monitoring.

8.2. TEST CASES

Test Case ID: UA_TC_001

- **Input:** Valid username and password
- **Expected Result:** User should be authenticated successfully and redirected to dashboard
- **Actual Result:** User logged in and dashboard loaded
- **Status:** Pass

Test Case ID: UA_TC_002

- **Input:** Invalid username or password
- **Expected Result:** Login should be denied with an error message
- **Actual Result:** Access denied and error message displayed
- **Status:** Pass

Test Case ID: DA_TC_001

- **Input:** Successful login credentials
- **Expected Result:** User should gain access to the dashboard
- **Actual Result:** Dashboard loaded successfully
- **Status:** Pass

Test Case ID: CH_TC_001

- **Input:** User selects "View Crime Hotspots"
- **Expected Result:** Crime hotspots should be displayed on the map with color-coded risk levels
- **Actual Result:** Map displayed with hotspots in Red, Orange, White, and Green
- **Status:** Pass

Test Case ID: RS_TC_001

- **Input:** User enters source and destination locations
- **Expected Result:** System should display multiple routes with risk assessment
- **Actual Result:** Routes displayed with safest route highlighted
- **Status:** Pass

Test Case ID: RRE_TC_001

- **Input:** Selected route from source to destination
- **Expected Result:** System should calculate risk score for each route
- **Actual Result:** Risk levels calculated and safest route recommended
- **Status:** Pass

Test Case ID: RSA_TC_001

- **Input:** User traveling along selected route
- **Expected Result:** System should send alert if entering a high-risk area
- **Actual Result:** Safety alert triggered successfully
- **Status:** Pass

Test Case ID: UL_TC_001

- **Input:** Click "Logout" button
- **Expected Result:** User session should terminate and redirect to login page
- **Actual Result:** Logout successful and redirected to login page
- **Status:** Pass

Test Case ID: AD_TC_001

- **Input:** Admin uploads latest crime data
- **Expected Result:** Data should be updated in the system and reflected on maps
- **Actual Result:** Data uploaded and updated successfully
- **Status:** Pass

Test Case ID: AN_TC_001

- **Input:** High-risk zone detected with multiple incidents
- **Expected Result:** Authorities should receive an alert
- **Actual Result:** Alert sent successfully to registered authorities
- **Status:** Pass

Test Case ID: MN_TC_001

- **Input:** User zooms or pans the crime hotspot map
- **Expected Result:** Map should adjust smoothly without losing hotspot accuracy

- **Actual Result:** Map navigation smooth and hotspots visible
- **Status:** Pass

Test Case ID: HT_TC_001

- **Input:** Admin requests crime trend analytics
- **Expected Result:** System should generate charts/graphs based on historical data
- **Actual Result:** Trend analysis displayed accurately
- **Status:** Pass

Test Case ID: ML_TC_001

- **Input:** Crime data fed to ST-GCN model
- **Expected Result:** Model should output risk probability for each zone
- **Actual Result:** Risk probabilities generated successfully
- **Status:** Pass

8.3. TEST REPORT

Introduction

The Test Report provides a comprehensive overview of the testing activities conducted for the project. The purpose of this report is to summarize testing outcomes, assess the system's functionality, reliability, and performance, and ensure that all modules operate correctly under real-world scenarios for women's safety and crime prediction.

Test Objective

The primary goal of testing was to validate the functionality, accuracy, and responsiveness of the predictive crime hotspot and route safety system. Specific objectives include:

- To verify user authentication and secure access to the platform.
- To validate real-time crime hotspot prediction and risk assessment accuracy.
- To assess route optimization and safe path recommendation based on risk levels.
- To evaluate the reliability of real-time alerts and notifications for high-risk zones.
- To ensure the ST-GCN model accurately predicts crime-prone areas.
- To ensure the system performs reliably under simultaneous multi-user requests.

Test Scope

Testing covered all critical components of the system:

- User login, registration, and authentication.

- Dashboard access and map visualization of crime hotspots.
- ST-GCN model performance and crime risk prediction.
- Safe route recommendation and route risk evaluation.
- Real-time alert notifications to users and authorities.
- Historical crime trend analysis and admin data updates.
- System response under concurrent users and multiple route simulations.

Test Environment

Hardware:

- Processor: Intel Core i5-11400 @ 2.6GHz
- RAM: 16GB DDR4
- Storage: 512GB SSD
- Network: Gigabit Ethernet

Software:

- Operating System: Windows 10 Pro
- Web Browser: Google Chrome, Mozilla Firefox
- Database: MySQL
- Web Server: WampServer 3.2.0
- Programming Language: Python 3.9
- Frameworks & Libraries: Flask, Bootstrap, Pandas, NumPy, Geopandas, Folium, TensorFlow/PyTorch

Bug Report

BID	**TCID**	**Bug Description**	**Status**	**Output**
BR_001	UA_TC_002	Invalid login credentials not showing proper error	Closed	Error message displayed: "Invalid credentials."
BR_002	RSA_TC_001	Alert notification delayed for high-risk zones	Closed	Notification timing optimized; alerts delivered in real-time
BR_003	RRE_TC_001	Risk score miscalculated for complex multi-route paths	Closed	Algorithm adjusted; accurate risk scores displayed

Test Conclusion

The testing phase demonstrates that project meets all functional and non-functional requirements. Modules including user authentication, ST-GCN-based crime hotspot prediction, route safety evaluation, real-time alerts, and map visualization performed as expected. Minor issues discovered during testing were resolved promptly. The system is ready for deployment, providing accurate, reliable, and real-time predictive safety support for women navigating high-risk urban areas.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENT

9.1. CONCLUSION

In conclusion, the proposed project is an effective tool for predicting high-risk crime zones and providing women with safer travel routes. The system leverages the Spatio-Temporal Graph Convolutional Network (ST-GCN) algorithm for real-time crime hotspot prediction and integrates Google Maps API for interactive visualization and route-based recommendations. The system is designed with multiple modules, including Crime Hotspot Analyser, Predictor, Map Visualization, Route Analysis, and Alert & Notification, each contributing to overall functionality and enhancing user safety. Compared to traditional manual methods and basic statistical models, the system provides more accurate, dynamic, and actionable predictions, allowing law enforcement agencies to allocate resources proactively and users to navigate safely. During testing and validation, the system demonstrated high reliability in predicting hotspots, assessing route safety, and generating timely alerts. The user-friendly interface and real-time notifications ensure an intuitive and responsive experience. Furthermore, the system has the potential for future expansion, including integration with emergency services, mobile alerts, real-time social media crime reporting, and coverage of additional regions or crime types. Overall, the project platform is a significant step towards empowering women, enhancing public safety, and supporting proactive crime prevention in urban environments.

9.2. FUTURE ENHANCEMENT

There are several areas where this can be improved in the future. Some of these areas are:

- **Integration with social media platforms:** The system can be integrated with social media platforms, such as Twitter and Facebook, to gather additional data on crime incidents and public perceptions of safety in certain areas.
- **Integration with emergency services:** The system can be integrated with emergency services, such as police and ambulance services, to enable faster response times to crime incidents in hotspots.
- **Expansion to other regions:** The system can be expanded to cover other regions beyond the current scope. This could involve integrating additional data sources and customizing the system to suit the specific needs of each region.

- **Integration with real-time crime data:** The system can be enhanced by integrating with real-time crime data from police departments or other crime reporting sources. This will help in improving the accuracy of the system and make it more effective in predicting crime hotspots.
- **Integration with mobile devices:** The system can be integrated with mobile devices to provide real-time alerts to women about crime hotspots in their vicinity. This will make the system more user-friendly and convenient to use.

Collaboration with law enforcement agencies: Collaboration with law enforcement agencies can help in improving the accuracy of the system and make it more effective in predicting crime hotspots. Law enforcement agencies can provide valuable insights and feedback on the accuracy of the system

CHAPTER 10
REFERENCES

1. Mohler, G. O., Short, M. B., Brantingham, P. J., Schoenberg, F. P., & Tita, G. E. (2011). Self-exciting point process modeling of crime. Journal of the American statistical Association, 106(493), 100-108.
2. Malik, O. M., Hussain, I., & Rehman, A. (2019). Predicting crime hotspots using deep learning. In 2019 International Conference on Innovative Computing (ICIC) (pp. 1-6). IEEE.
3. Gerber, M. S., Johnson, M. P., & Berman, M. (2011). Predictive policing and reasonable suspicion. Emory Law Journal, 60, 585.
4. Liu, Y., & Jiang, S. (2019). Crime prediction using data mining techniques: A comprehensive review. IEEE Transactions on Cybernetics, 49(4), 1249-1269.
5. Ratcliffe, J. H., & Rengert, G. F. (2008). Near repeat patterns in Philadelphia shootings. Security Journal, 21(1-2), 23-40.
6. Bowers, K. J., & Johnson, S. D. (2005). Predictive mapping of crime by ProMap: Accuracy, units of analysis, and the problem of vagueness. In Crime mapping: New tools for law enforcement (pp. 221-248). Taylor & Francis.
7. Chainey, S., Tompson, L., & Uhlig, S. (2008). The utility of hotspot mapping for predicting spatial patterns of crime. Security Journal, 21(1-2), 4-28.
8. Li, X., Liu, L., Lv, T., & Cheng, T. (2017). Crime hotspot detection using spatial filtering and density peaks clustering. Journal of Ambient Intelligence and Humanized Computing, 8(6), 905-914.
9. Wang, Y., Chen, L., & Zhang, J. (2019). A novel approach for spatio-temporal prediction of crime hotspots based on deep learning and multi-source data fusion. Information Fusion, 47, 60-68.
10. Wu, Y., & Leung, Y. (2018). Crime hotspot prediction using deep learning. In Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (pp. 2354-2363).
11. Badea, A., Popescu, D., & Antonie, M. L. (2017). Crime prediction using spatio-temporal clustering and SVM classification. In 2017 IEEE 13th International Conference on Intelligent Computer Communication and Processing (ICCP) (pp. 227-234). IEEE.

12. Ratcliffe, J. H. (2010). The spatial extent of criminogenic places: A changepoint regression of violence around bars. Geographical Analysis, 42(3), 215-231.
13. Ratcliffe, J. H., Taniguchi, T., & Groff, E. R. (2011). Mapping the criminogenic environment: Hot spots analysis using kernel density estimation. In Place matters (pp. 67-97). Springer, New York, NY.
14. Zhu, H., Xiong, H., Shen, H. T., & Cheng, X. (2014). Discovering spatio-temporal crime hotspots through exploratory space-time analysis. In 2014 IEEE 30th International Conference on Data Engineering (pp. 880-891). IEEE.

BOOK REFERENCES

1. Ahlstrom, C. & Allen, T. (2014). Applying predictive analytics in the public sector: Techniques and examples. Boca Raton: CRC Press.
2. Bhattacharyya, S., Kalita, J. K., & Sarma, K. K. (2014). Crime modeling and mapping using geospatial technologies. Boca Raton: CRC Press.
3. Chang, K. M. (2016). Data-Driven Crime Prevention: With Social Network Analysis and Simulation Technologies. Singapore: Springer.
4. Huang, H., & Zhang, W. (2015). Mining spatio-temporal patterns and trajectory in crime data mining and visualization. Boca Raton: CRC Press.
5. Ratcliffe, J. H. (2010). Intelligence-led policing. London: Routledge.
6. Ratcliffe, J. H. (2012). Spatial analysis for the social sciences. New York: Springer.
7. Santos, R., & Cheng, T. (2017). Intelligent techniques in engineering management: Theory and applications. Boca Raton: CRC Press.
8. Wang, J., Zhang, J., & Zhang, C. (2014). Mining spatio-temporal patterns and trajectory in crime data mining and visualization. Boca Raton: CRC Press.

WEB REFERENCES

1. "Python Flask Tutorial." Flask Tutorial, www.tutorialspoint.com/flask/index.htm.
2. "Google Maps Platform." Google Cloud Platform, cloud.google.com/maps-platform/.
3. "Support Vector Machines." Scikit-learn, scikit-learn.org/stable/modules/svm.html.
4. "MySQL :: MySQL Connector/Python Developer Guide." MySQL Developer Zone, dev.mysql.com/doc/connector-python/en/.
5. Jadhav, J., & Darade, T. (2021). Women's Safety using Machine Learning: A Review. International Journal of Advanced Research in Computer Science, 12(1), 1-5.